frontiers in Education

TYPE Systematic Review
PUBLISHED 05 August 2026
DOI 10.3389/feduc.2026.1902490

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OPEN ACCESS

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RECEIVED 07 June 2026
REVISED 09 July 2026
ACCEPTED 21 July 2026
PUBLISHED 05 August 2026

CITATION
García-Holgado L, García-Peñalvo FJ and García-Holgado A (2026) Technology-mediated cultural inclusion and diversity initiatives in primary and secondary education: a systematic literature review.
Front. Educ. 11:1902490.
doi: 10.3389/feduc.2026.1902490

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Technology-mediated cultural inclusion and diversity initiatives in primary and secondary education: a systematic literature review

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Cultural diversity and inclusion have become central challenges in contemporary education, particularly in multicultural societies where primary and secondary schools play a crucial role in fostering intercultural awareness. Technology has increasingly been employed as a mediator to support inclusive educational practices, yet the existing evidence remains scattered and heterogeneous. This study presents a systematic literature review (SLR) to identify, classify, and synthesize technology-mediated initiatives that address cultural inclusion and diversity in primary and secondary education, with particular attention to the evidence that can inform school leadership, technology adoption, and institutional decision-making. In accordance with PRISMA guidelines, we conducted a systematic search across major databases, applying predefined inclusion and exclusion criteria. A total of 47 studies were assessed for eligibility, yielding 28 distinct initiatives. The analysis combined quantitative mapping of initiative types, technologies, and evaluation mechanisms with qualitative synthesis of challenges and facilitators. The findings show that most initiatives rely on accessible technologies such as digital platforms, collaborative tools, and content creation applications to foster intercultural understanding. However, evaluation practices are often limited, with a lack of longitudinal and rigorous designs. The results also highlight persistent challenges, including insufficient teacher training, infrastructural inequalities, and institutional constraints. At the same time, technology demonstrates significant potential to enable cross-cultural interaction, enhance accessibility, and support inclusive pedagogical strategies. This review provides an integrated overview of current practices, identifies decision-relevant research gaps, and offers implications for educational leaders seeking to implement, evaluate, and sustain equitable technology-mediated cultural inclusion initiatives in schools.

KEYWORDS
cultural diversity, cultural inclusion, educational leadership, ICT in education, K-12, primary education, secondary education, systematic review

1 Introduction

In recent decades, schools have become increasingly diverse spaces where students from different cultural, linguistic, and socio-economic backgrounds converge. Promoting cultural inclusion and diversity in education is not only a matter of social justice but also a key factor in fostering democratic citizenship and intercultural understanding. Primary and secondary schools, in particular, serve as foundational spaces where young learners develop attitudes and skills necessary for living in plural societies.

In this article, the term cultural diversity refers to the presence of students who differ in language, ethnicity, nationality, religion and migration histories within the same school community. Cultural inclusion goes beyond the mere recognition of this diversity to ensure that all students participate equitably in learning and school life, and that their identities are valued rather than assimilated. This understanding is rooted in the tradition of multicultural education, which calls for transforming curricula, pedagogy and school culture to promote equity, prejudice reduction and an empowering school environment (Banks, 2014; Nieto and Bode, 2012). Intercultural education and recent work on intercultural competence, emphasizes respectful dialogue, reciprocal learning and the development of skills to interact across differences (Dervin and Gross, 2016; UNESCO, 2006). Closely related are culturally responsive and culturally relevant pedagogies, which use learners' cultural backgrounds, experiences and ways of knowing as resources for teaching, and aim to foster academic success, cultural competence and critical consciousness (Gay, 2002; Ladson-Billings, 1995). This review adopts these intertwined perspectives to focus specifically on technology-mediated initiatives that address cultural diversity and inclusion in primary and secondary education.

While numerous educational initiatives have sought to address cultural inclusion (Banks, 2014; Nieto and Bode, 2012; UNESCO, 2006), technology has emerged as a powerful mediator, enabling innovative forms of collaboration, communication, and learning. For school leaders, this fragmentation creates a practical challenge: decisions about which technologies to adopt, how to support teachers, and how to sustain culturally inclusive innovation are often made without a consolidated evidence base. Existing studies are often fragmented, as they tend to examine isolated interventions, specific technologies, or context-specific case studies rather than synthesizing the field as a whole. For example, Morales-Chicas et al. (2019) reviewed culturally relevant computing education, Marinagi et al. (2017) focused on technology for students with disabilities, and Solís-Pinilla et al. (2024) examined educational interactions within the broader framework of school culture. To date, no comprehensive systematic literature review has synthesized the available evidence on how technology is being used to support cultural diversity and inclusion in primary and secondary education.

Several prior reviews illustrate this fragmentation. Morales-Chicas et al. (2019) focused narrowly on computing education within culturally relevant pedagogies, whilst other reviews have addressed broader inclusion concepts without the cultural specificity required. For instance, recent systematic reviews have examined technology for students with disabilities or general inclusive education practices, but these predominantly focus on special educational needs rather than cultural diversity (Marinagi et al., 2017). Lunn et al. (2022) explored empathy development in STEM fields in higher education and Sudirman (2019) focused on teaching innovations for vocational teachers, without addressing cultural inclusion. Solís-Pinilla et al. (2024) addressed educational interactions, including multiculturality, within the broader framework of school culture rather than specifically technology-mediated cultural inclusion.

However, studies that have addressed cultural aspects have employed different scopes or methodologies that do not align with the specific focus of this review. Snijder et al. (2021) examined culturally sensitive wellbeing and substance use prevention programs for Indigenous populations, although this was an empirical study rather than a systematic literature synthesis. De Prado and Gago (2020) explored cultural inclusion through museum-school collaborations, though this focused on accessibility for functional diversity rather than cultural diversity *per se*. Similarly, whilst Pandey et al. (2023) developing inclusive computing curricula, this concentrated on making computer science accessible rather than addressing cultural inclusion directly.

This fragmentation highlights the need for a systematic review that integrates findings, identifies patterns, and uncovers gaps in the literature.

This is the first PRISMA-aligned systematic review to synthesize technology-mediated initiatives specifically targeting cultural diversity and inclusion in primary and secondary (K-12) education for the period 2014–2026. This study systematically reviews and analyses these initiatives. Although the review is not limited to leadership studies, its synthesis is intended to provide decision-relevant evidence for school leaders and other stakeholders involved in implementing and sustaining technology-mediated inclusion initiatives. To guide this review, the following research questions were formulated:

- RQ1: What types of initiatives are being implemented to promote cultural inclusion and diversity through technology in primary and secondary education?
- RQ2: What mechanisms are used to evaluate the impact of these initiatives?
- RQ3: What are the main challenges and barriers faced by schools in implementing cultural inclusion and diversity projects?
- RQ4: Which key affordances of digital technologies are used in cultural-inclusion initiatives in primary and secondary education, and in which contexts are they reported?

The article is structured as follows. Section 2 outlines the methodology of the systematic literature review, including inclusion/exclusion criteria, databases, and the selection process. Section 3 presents the mapping results, providing a classification of the initiatives identified. Section 4 synthesizes the results in relation to the research questions, followed by Section 5, which discusses their implications and biases. Finally, Section 6 concludes with a summary of findings, contributions, limitations, and recommendations for future research.

2 Methods

This study followed the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) in its 2020 version (Page et al., 2021) to ensure transparency, reproducibility, and rigor in the review process. A systematic literature review (SLR) and an accompanying mapping component were conducted to identify, select, and synthesize research on technology-mediated initiatives to promote cultural inclusion and diversity in primary and secondary education. The review focused on descriptive and explanatory synthesis of heterogeneous educational interventions rather than on quantitative meta-analysis.

A review protocol specifying the research questions, eligibility criteria, information sources, search strategy, screening procedures, data extraction, and quality appraisal was developed *a priori* by the research team and piloted on a small subset of studies. Although the protocol was not prospectively registered in an open repository (e.g., the Open Science Framework), it guided the review process from study identification to data synthesis and was applied consistently throughout all search updates.

To define the research questions and the search strategy, the PICOC framework was followed to guide and delimit the population, intervention, comparison, results, and context as shown in Table 1.

2.1 Inclusion and exclusion criteria

The inclusion and exclusion criteria were designed to ensure the relevance, quality, and accessibility of the studies selected for this systematic review. Each criterion is aligned with the research question and the PICOC framework, providing transparency and replicability to the selection process.

The selected studies had to meet at least the following criteria:

- IC1: Initiatives that work on cultural inclusion and/or cultural diversity in primary and secondary education.
- IC2: Technology-mediated initiatives.
- IC3: Publications written in English or Spanish.
- IC4: Publications that are peer-reviewed journal articles, conference papers, books, and book chapters that have undergone a peer-reviewed process and are not literature reviews.
- IC5: The publication is the most recent and comprehensive of the related publications on the same study.
- IC6: Publication accessible in full text.

TABLE 1 PICOC framework.

Element	Description
P (Population)	Educational initiatives.
I (Intervention)	Working on technology-mediated cultural inclusion and diversity.
C (Comparison)	No comparator was required.
O (Outcome)	Educational initiatives that address technology-mediated cultural inclusion and diversity implemented in real-life contexts.
C (Context)	Primary and secondary education.

Studies that met any of the following conditions were excluded from the review:

- EC1: Initiatives that do not address cultural inclusion or cultural diversity in primary and secondary education.
- EC2: Initiatives that are not mediated by technology.
- EC3: Publications not written in English or Spanish.
- EC4: Publications other than journal articles, conference papers, edited books or book chapters; works that have not undergone peer review; literature reviews.
- EC5: The publication is not the most recent and comprehensive of related publications on the same study.
- EC6: The publication is not accessible in full text.

2.2 Search strategy

A systematic search was conducted in two multidisciplinary bibliographic electronic databases: Scopus (Elsevier) and Web of Science (Clarivate). They were selected because they are widely recognized as comprehensive, high-quality citation indexes that cover peer-reviewed journals, books and conference proceedings across education, social sciences and technology, and because their advanced search options support transparent, reproducible Boolean strategies required for systematic reviews. Comparative evaluations of academic search systems identify Scopus and Web of Science as among the few databases that consistently meet minimum standards of precision, recall and reproducibility for evidence syntheses, whereas tools such as Google Scholar are not recommended as primary search systems (Gusenbauer and Haddaway, 2020; Martín-Martín et al., 2021). Their combined use is widely recommended to maximize retrieval coverage and reduce database-specific bias in evidence syntheses. The review covered publications from 2014 to 2026. The original search was conducted for the period 2014–2024, and two update searches were subsequently performed in 2025 and 2026 using the same conceptual search blocks and adapted date limits. The time span was selected to ensure that the review reflects the most recent and relevant developments in the use of technology for cultural inclusion and diversity in education. The starting year coincides with a period of rapid expansion in technology-mediated educational initiatives, marked by increased adoption of digital platforms, mobile learning, and inclusive education policies.

In both databases, the search strategy combined three main concept blocks related to (a) cultural inclusion and diversity, (b) technology, and (c) primary and secondary education, using Boolean operators.

The canonical search string was structured into three concept blocks combined with AND, as follows:

> ("cultural inclusion" OR "cultural inclusivity" OR "cultural diversity" OR multiculturalism OR multiculturality OR interculturalism OR interculturality OR transculturalism OR "cross-cultural education" OR "transcultural education" OR "culturally responsive teaching" OR "inclusive curriculum" OR "diverse learning environment" OR "cultural awareness")

> (ICT OR "information and communication technolog*" OR "information technolog*" OR "educational technolog*" OR technolog*)

> ("primary education" OR "secondary education" OR school OR K-12 OR baccalaureate OR "high school" OR "upper secondary education" OR "vocational training" OR "vocational education and training" OR VET)

The full search strings were adapted to the specific syntax of each database and were applied to the title, abstract and keyword/topic fields. In the original search, limits were set to publications between 2014 and 2024, written in English or Spanish, and classified as journal articles, conference papers, books, book chapters or review articles in education-related subject areas. In the update searches, the same strategy was rerun with adapted date limits to include newly published records up to 2026. The search terms have been adapted to the advanced search operators of Web of Science and Scopus:

Scopus:

> TITLE-ABS-KEY(education AND (ICT OR "information and communication technolog*" OR "information technolog*" OR "educational technolog*" OR technolog*) AND ("cultural inclusion" OR "cultural inclusivity" OR "cultural diversity" OR multiculturalism OR multiculturality OR interculturalism OR interculturality OR transculturalism OR "cross-cultural education" OR "transcultural education" OR "culturally responsive teaching" OR "inclusive curriculum" OR "diverse learning environment" OR "cultural awareness") AND ("primary education" OR "secondary education" OR school OR K-12 OR baccalaureate OR "high school" OR "upper secondary education" OR "vocational training" OR "vocational education and training" OR VET)) AND PUBYEAR > 2013 AND PUBYEAR < 2025 AND (LIMIT-TO (DOCTYPE,"ar") OR LIMIT-TO (DOCTYPE,"ch") OR LIMIT-TO (DOCTYPE,"bk") OR LIMIT-TO (DOCTYPE,"cp") OR LIMIT-TO (DOCTYPE,"re")) AND (LIMIT-TO (LANGUAGE,"English") OR LIMIT-TO (LANGUAGE,"Spanish"))

Web of Science:

> TS = ((education) AND (ICT OR "information and communication technolog*" OR "information technolog*" OR "educational technolog*" OR technolog*) AND ("primary education" OR "secondary education" OR school OR K-12 OR baccalaureate OR "high school" OR "upper secondary education" OR "vocational training" OR "vocational education and training" OR VET) AND ("cultural inclusion" OR "cultural inclusivity" OR "cultural diversity" OR multiculturalism OR multiculturality OR interculturalism OR interculturality OR transculturalism OR "cross-cultural education" OR "transcultural education" OR "culturally responsive teaching" OR "inclusive curriculum" OR "diverse learning environment" OR "cultural awareness"))

In Web of Science Core Collection, the Topic field (TS) was searched using the string above. Filters were then applied to restrict results to the years 2014–2024 (PY = 2014–2024), to documents in English or Spanish, and to the document types of articles, proceeding papers, books, book chapters, early access or review articles.

For transparency, the update searches conducted in 2025 and 2026 used the same conceptual search blocks and database-specific syntax as the original search, modifying only the publication-year limits. In Scopus, the update searches replaced the original year limit with the corresponding extended limits:

> update: PUBYEAR > 2023 AND PUBYEAR < 2026
> update: PUBYEAR > 2024 AND PUBYEAR < 2027

In Web of Science Core Collection, the Topic search string remained unchanged, and the year filters were updated as follows:

> update: PY = 2024–2025
> update: PY = 2025–2026

No additional conceptual terms were added in the update searches.

The original search was conducted in August 2024. Two update searches were conducted in August 2025 and April 2026. These updates used the same conceptual search blocks and database-specific syntax, with the publication-year limits extended to capture records published after the original search. Therefore, the final review period was 2014–2026.

2.3 Selection process

This study followed the PRISMA guidelines to ensure a rigorous and transparent examination of the literature. In line with these guidelines, a flow diagram (Figure 1) illustrates the stages of the review process, including identification, screening, eligibility assessment, and final selection of studies.

Three members of the research team independently screened all titles and abstracts against the inclusion and exclusion criteria. Records judged as potentially relevant by at least one reviewer were retrieved in full text. The same reviewers then independently assessed the full texts for eligibility and agreed on final inclusion. Any disagreements at either stage were resolved through discussion until a consensus was reached. No formal inter-rater reliability statistic, such as Cohen's kappa, was calculated, because the screening process was designed as an independent consensus-based procedure.

Data extraction was carried out by one reviewer using a piloted data extraction form that captured key characteristics of each initiative (context, participants, educational level, technology, type of diversity targeted and main outcomes).

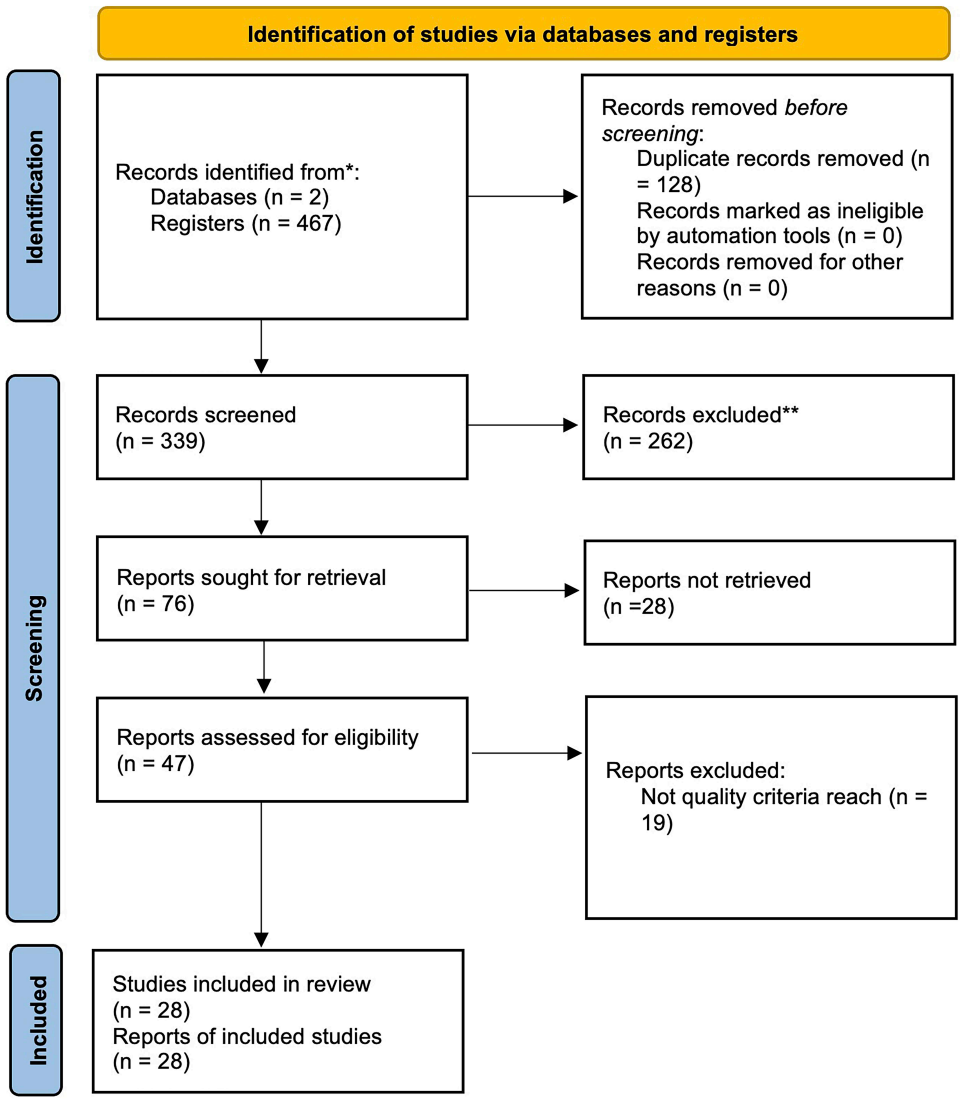


FIGURE 1
PRISMA workflow.

A second reviewer checked the extracted information for completeness and accuracy, and any issues were resolved through discussion.

The initial search (open database available at https://doi.org/10.5281/zenodo.20561193) conducted on 15 August 2024 identified 283 records. Two subsequent searches using the same strategies were carried out on 13 August 2025 and 19 April 2026, yielding a further 93 and 91 records respectively. Overall, the cumulative searches up to May 2026 identified 467 records, from which 128 duplicates were removed before screening. The remaining 339 records were screened at title and abstract level.

Most of the excluded studies were removed because they did not meet the core focus of the review (EC1), as they addressed higher education, preschool, adult learners, or non-educational contexts, rather than cultural inclusion or diversity in primary and secondary education. Others were excluded for focusing on related but different topics, such as gender in STEM, disability inclusion, cyberbullying, digital competencies, or general inclusion without a cultural dimension. A significant number of studies were also discarded because they did not constitute concrete initiatives (but rather theoretical analyses, teacher perception studies, or preliminary reports) or because they were not mediated by technology (EC2). Additional exclusions corresponded to literature reviews or secondary studies (EC4), duplicates where a more complete version was available (EC5), or works that were not accessible in full text (EC6). Overall, the exclusions ensured that only relevant, empirical, technology-mediated initiatives directly related to cultural inclusion and diversity in primary and secondary education were retained.

Finally, 76 full texts were sought, and in the eligibility stage, 47 studies were assessed for final eligibility. At this stage, quality criteria were also applied, including clarity of objectives, methodological transparency, and relevance to the research questions. A verification checklist was designed to evaluate the relevant aspects of the selected articles using a three-valued metric (García-Peñalvo, 2022): Yes (1), partial (0.5), or no (0) as shown in Table 2. Given the methodological heterogeneity of the included studies, we did not apply a single standardized

TABLE 2 Quality criteria.

Question	Score
1. Is the intervention designed to address diversity and inclusion in the classroom?	Yes/Partial/No
2. Was the proposal implemented in a real context?	Yes/Partial/No
3. Was the initiative implemented in more than one educational institution or educational level? Is it replicable?	Yes/Partial/No
4. Is the initiative sustainable over time?	Yes/Partial/No
5. Does the work integrate the use of technology to support the implementation of the initiative?	Yes/Partial/No
6. Does the proposal include a mechanism for evaluating its impact in the short, medium, or long term?	Yes/Partial/No
7. Does it identify limitations and propose future research directions related to the initiative?	Yes/Partial/No

appraisal tool such as MMAT, CASP, JBI, ROBINS-I, or RoB 2. Instead, we used a review-specific quality assessment checklist focused on seven methodological and reporting criteria aligned with the objectives of this review. This decision allowed us to assess consistently whether each study described a real technology-mediated initiative, addressed cultural inclusion or diversity, reported implementation details, included an evaluation mechanism, and discussed limitations or future directions. To ensure the quality of the evidence, a minimum inclusion threshold of 4.5 points was set on this scale, reflecting that most criteria are fully met and that there are no serious shortcomings. This threshold corresponds to approximately 65% of the maximum score and was selected as a conservative cut-off for this review, rather than as a universally accepted standard (Hoffmann-Eßer et al., 2018). The initial search in August 2024 led to the inclusion of 16 studies, the update in August 2025 contributed 5 additional studies, and the April 2026 update added a further 7, giving a total of 28 studies included in the review.

The data synthesis process was carried out in two steps: (1) a quantitative mapping to classify the initiatives and their evaluation mechanisms, and (2) a qualitative synthesis to address the research questions, identify emerging themes, and highlight gaps in the literature. The results of these analyses are presented in the following section.

3 Results

3.1 Overview of the selected studies

The final dataset used in this study consists of 28 primary studies selected after applying the inclusion and exclusion criteria defined in the review protocol. To facilitate traceability and support a structured presentation of the evidence, each paper was assigned a unique identifier. Table 3 presents the complete dataset, including the ID assigned to each paper, its author, and its corresponding title.

This dataset constitutes the empirical basis for the subsequent analysis. After presenting the final set of selected studies, the dataset is further characterized through a systematic mapping process. Specifically, we examine the studies according to the mapping techniques associated with the seven mapping questions defined in this review. These questions guide the classification and interpretation of the selected papers, allowing us to identify publication trends, research contexts, methodological approaches, application domains, and other relevant dimensions of the literature.

3.1.1 Geographical distribution of the initiatives

The initiatives have been deployed across a wide geographical area spanning several continents. Studies involving more than one country or continent are counted in each of the relevant categories; therefore, the total number of country–continent occurrences is higher than the total number of studies (see Table 4).

The studies were conducted across four continents and in 20 countries, as shown in Figure 2. In Europe, the studies included Ireland, the United Kingdom (2), Italy, Russia, and Ukraine, for a total of six studies. In America, the studies included the United States (4), Argentina, Colombia (2), Chile (2), Ecuador, France (Guadeloupe, French overseas territory in the Caribbean), and Canada, for a total of 12 studies. In Asia, the studies included Taiwan, India, Qatar, South Korea, Indonesia (5), Thailand (2), and Japan (2), for a total of thirteen studies. In Oceania, the studies included New Zealand and the United States (Hawaii), for a total of two studies. No studies were conducted in Africa or Antarctica.

3.1.2 Publication trends over time

Figure 3 shows the distribution of studies by publication year. The results indicate an irregular publication trend between 2014 and 2026, with relatively few studies published in the earlier years. A noticeable increase is observed from 2024 onward, reaching a peak in 2025 with 8 studies. This suggests a growing research interest in the topic in recent years.

3.1.3 Temporal evolution of the initiatives

Table 5 shows a broad temporal distribution of the initiatives, spanning from the late 1990s to 2026. The earliest references correspond to 1999–2012 and 2009–2014, while most initiatives are concentrated from the mid-2010s onward. A limited number of initiatives started between 2010 and 2014, followed by a clear increase during 2015–2019 and 2020–2024, with eight initiatives in each period. The most recent period, 2025–2026, also shows a high concentration, with seven initiatives, most of them starting in 2025.

Periods were defined according to the starting year of each initiative, using five-year intervals to facilitate comparison across time while keeping earlier initiatives grouped separately due to their limited frequency. When grouping initiatives by starting year into broader periods, the distribution confirms this recent concentration. Only two initiatives began before 2010, while a limited number started in the early 2010s. Activity increases

TABLE 3 Selected studies included in the systematic literature review.

Study ID	Authors	Title
P001	Rickard et al. (2014)	Assessing Impact of ICT Intercultural Work
P002	Hopper (2014)	Bringing the World to the Classroom Through Videoconferencing and Project-Based Learning
P003	Cheng et al. (2021)	Culturally Responsive Teaching in Technology-Supported Learning Environments in Marine Education for Sustainable Development
P004	Baldivieso and Carrasco (2014)	Digital Public School in Native Peoples: University of La Punta's Proposal for Ranquel and Huarpe Communities in San Luis Province, Argentina
P005	Sarangapani et al. (2019)	Fostering Deep Learning in Cross-Cultural Education Through the Use of Content-Creation Tools
P006	Garzón-Díaz (2021)	From Cultural Awareness to Scientific Citizenship: Implementing Content and Language Integrated Learning Projects to Connect Environmental Science and English in a State School in Colombia
P007	Liesaputra et al. (2020)	Future-Proofing Kiwi Kids Through the Use of Digital Technology
P008	Quintero Solano et al. (2019)	Intercultural Effectiveness and Video Use: Characterization of Basic High School Students' Audiovisual Production from the Colombian Caribbean
P009	Careaga Butter et al. (2015)	Intercultural Talent Management Model: Virtual Communities to Promote Collaborative Learning in Indigenous Contexts: Teachers' and Students' Perceptions
P010	Estrada et al. (2024)	Multiculturalism and Teacher Training in Intercultural Bilingual Education
P011	Barzanò et al. (2017)	New Means and New Meanings for Multicultural Education in a Global–Italian Context
P012	Katbeh et al. (2023)	Promoting STEM Education Through the Preparation of Multicultural National Robotics Teams in Qatar: Evaluation
P013	Seo and Lim (2024)	Racial Literacy Development in Online Intercultural Education in the EFL Classroom
P014	Huff-Lohmeier et al. (2017)	Research in Optics for K–14 Teachers (ROKET): A Research Experience for Teachers in Native American Schools
P015	Rahmawati et al. (2020)	The Integration of Ethnopedagogy in Science Learning to Improve Student Engagement and Cultural Awareness
P016	Gubanihina et al. (2017)	The Pedagogical Potential of Information and Communication Technologies in the Formation of Ethnic and Cultural Awareness of Younger Schoolchildren
P017	Bowman et al. (2024)	Impact of Professional Development in Culturally Relevant Engineering Design for Elementary and Middle School Teachers (RTP, Diversity)
P018	Christensen and Knezek (2025)	Impact of Teaching Simulations on Resilience, Empathy, and Culturally Responsive Teaching Self-Efficacy in Technology Teacher Education
P019	Patras et al. (2025)	Integration of Culturally Responsive Teaching Approach, Local Wisdom, and Gamification in Pancasila Education to Develop Students' Multicultural Competence
P020	Parent et al. (2025)	Integration of Interculturality in Education for Sustainable Development
P021	Mutchima et al. (2025)	The Effect of Metaverse Technology on Multicultural Learning: Strengthening the Social Attitudes, Cultural Awareness, and Critical Thinking Skills of Secondary School Students
P022	Pattaufi et al. (2025)	Developing Augmented Reality as a Teaching Material to Enhance Cultural Awareness in Secondary Schools
P023	Pamungkas et al. (2025)	Fostering Nationalism Through Digital Poetry: A Design-Based Study
P024	Nisa et al. (2025)	Integrating Culturally Responsive Teaching and TPACK Frameworks in the Design and Validation of Indonesian Language Learning Tools for Multicultural Schools
P025	Sugiyama et al. (2026)	Integrating Japanese Festival Traditions into Computational Thinking: A Culturally Responsive Approach to Micro:bit-Based Physical Computing
P026	Gonzalez-Vidal (2026)	Navigating Linguacultures Through Fanfiction: Unveiling Opportunities and Challenges in the Chilean EFL School Context
P027	Ostrovski et al. (2026)	Original Digital Technologies and Artificial Intelligence for Multicultural Competence in High School Literature
P028	Mayuroh et al. (2025)	The Impact of Internet-Based International Small-Group Activities on Students in the Countryside

TABLE 4 Distribution of studies by country and continent.

Study ID	Countries	Continents
P001	Ireland, United Kingdom (Northern Ireland)	Europe
P002	United States	America
P003	Taiwan	Asia
P004	Argentina	America
P005	India, United Kingdom	Asia/Europe
P006	Colombia	America
P007	New Zealand	Oceania
P008	Colombia	America
P009	Chile	America
P010	Ecuador	America
P011	Italy	Europe
P012	Qatar	Asia
P013	South Korea, United States (Hawaii)	Asia/Oceania
P014	United States	America
P015	Indonesia	Asia
P016	Russia (Nizhny Novgorod)	Europe
P017	United States	America
P018	United States	America
P019	Indonesia	Asia
P020	France (Guadelupe), Canada	America
P021	Thailand	Asia
P022	Indonesia	Asia
P023	Indonesia	Asia
P024	Indonesia	Asia
P025	Japan	Asia
P026	Chile	America
P027	Ukraine	Europe
P028	Thailand and Japan	Asia
TOTAL	33	30

substantially from 2015 onward and remains consistently high in the 2020s.

Overall, the data suggest that although some initiatives began earlier and span longer periods, most initiatives are concentrated in recent years, particularly from 2020 onward.

3.1.4 Authorship and institutional patterns

The analysis of authorship shows a highly dispersed research community. A total of 99 unique authors were identified across the selected studies. Rahmawati Y is the only author appearing in more than one publication, who appears in two studies. In contrast, all other authors contributed to a single study, indicating the absence of a clearly dominant research group in this area.

Across the included studies, the mean number of authors per paper is 3.57 and the median is 4. Only 3 studies are single-authored (10.7%), whereas 25 studies involve multiple authors (89.3%), showing a strong preference for collaborative publication.

Considering the normalized affiliations and their association with authors, the most represented institutions are Universitas Negeri Jakarta and the National Research and Innovation Agency (BRIN), each with six author appearances. Other highly represented affiliations include Universidad del Norte, Lobachevsky State University of Nizhny Novgorod (Arzamas Branch), and Suan Dusit University, each with five author appearances. Overall, the data suggest a geographically diverse institutional landscape, with particularly strong representation from Indonesian institutions, followed by contributions from Latin America, Europe, Asia, and North America.

3.1.5 Publication venues and document types

The analysis of publication types shows that journal articles are the dominant publication format, as shown in Table 6, accounting for 22 out of 28 studies (78.6%). In contrast, conference papers represent 6 publications (21.4%). This distribution indicates that research in this area is primarily disseminated through peer-reviewed academic journals, with conference venues playing a secondary role.

3.1.6 Technologies used to support cultural inclusion and diversity

The technological landscape of the selected studies ($n = 28$) was organized into five macro-categories according to their pedagogical function: (i) learning management and institutional platforms; (ii) synchronous and asynchronous communication/collaboration tools; (iii) content-creation and interactive learning tools; (iv) physical computing, robotics and STEM toolkits; and (v) immersive and simulation-based environments (VR, metaverse, virtual tours).

Each study was assigned to a single primary technology category based on the environment that structured most of the learning activities in the intervention (see Table 7). When multiple technologies were used, we classified the study under the category most central to the pedagogical design and recorded additional tools as secondary descriptors. For example, in the case of the study P027, it used Artificial Intelligence tools and incorporated communication platforms, so it was classified under content-creation and interactive learning tools because AI-based applications were primarily used to generate, transform, and support engagement with textual, visual, and audiovisual literary content.

As shown in Table 8, the most frequently reported technologies are content-creation and interactive learning tools ($n = 9$), followed by synchronous and asynchronous communication/collaboration tools ($n = 6$). Learning management and institutional platforms ($n = 4$) and physical computing, robotics and STEM toolkits ($n = 5$) are also commonly used to support learning activities. In contrast, immersive and simulation-based environments ($n = 4$) appear less frequently across the reviewed studies. Overall, the distribution suggests that research on cultural inclusion and diversity in primary and secondary education primarily relies on widely accessible digital tools for communication, content creation and collaboration, while more advanced or immersive technologies remain comparatively underrepresented.

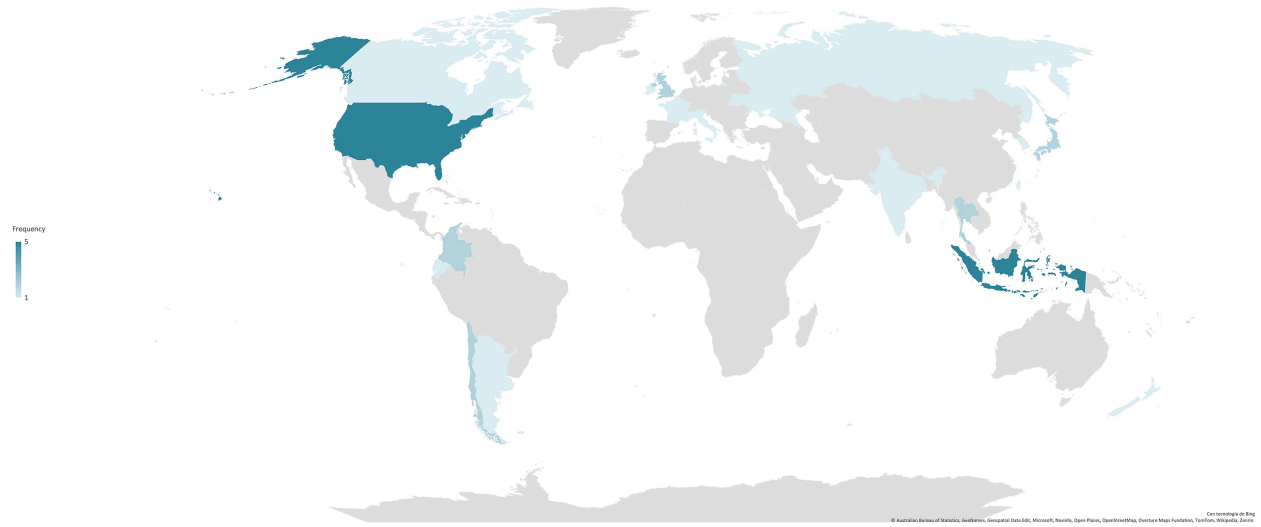


FIGURE 2
Countries distribution.

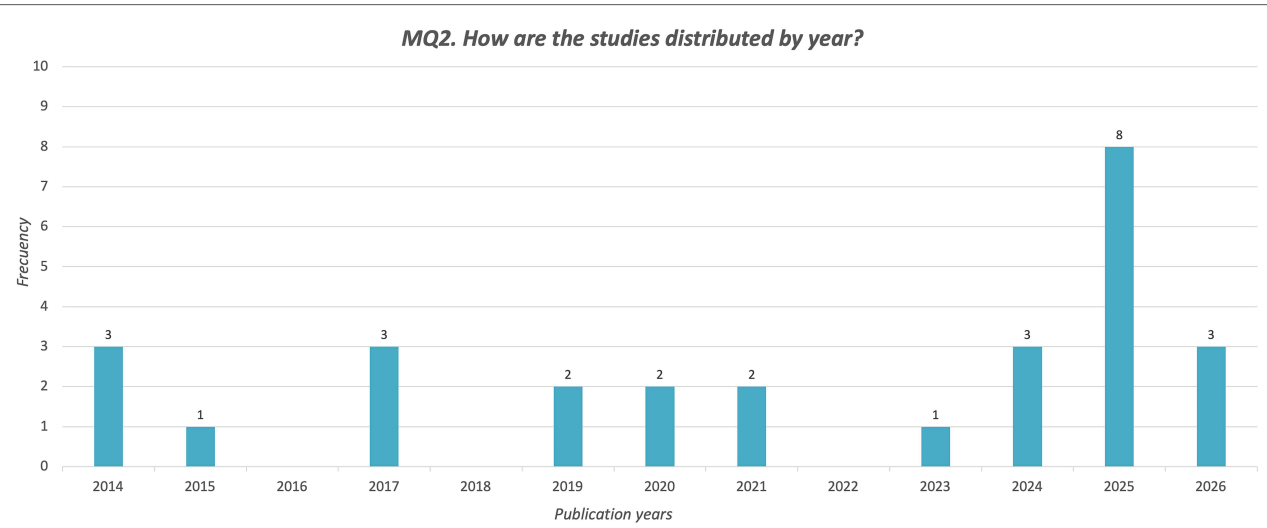


FIGURE 3
Studies distribution.

TABLE 5 Initiatives classification by year.

Period	Frequency	Examples
Before 2010	2	1999–2012, 2009–2014
2010–2014	3	2010s, 2010–2016, 2012–2013
2015–2019	8	2015–2017, 2017, 2017–2018, 2017–2020, 2018, 2018–2019, 2019, 2019–2020
2020–2024	8	2022, 2022–2023 (2), 2023–2024 (3), 2024 (2)
2025–2026	7	2025 (6), 2026

3.1.7 Forms of cultural inclusion and diversity addressed

The five categories were selected to capture cultural diversity from complementary dimensions, including general awareness, structural inequalities (ethnic and Indigenous diversity), interaction processes (interculturality), communicative factors (linguistic diversity), and mobility-related dynamics (migration). To select the thematic categories, an iterative qualitative synthesis was conducted. First, each study was coded according to the main dimension of cultural inclusion or diversity explicitly addressed by the authors. Second, similar codes were grouped into broader thematic categories through comparison

across studies. Third, the categories were reviewed by the research team to ensure conceptual consistency and alignment with the research questions. This process led to five thematic categories: multicultural awareness, ethnic and Indigenous diversity, intercultural and cross-cultural education, linguistic diversity, and migration-related dynamics. These categories were not defined *a priori*, but emerged from the dominant focus of the included initiatives during the synthesis process. Together, these categories provide a comprehensive yet analytically manageable framework to examine how inclusion and cultural diversity are addressed across different contexts.

In this review, general awareness refers to broad institutional or pedagogical commitments to inclusion and diversity without a specific target group; ethnic and Indigenous diversity focuses on structural inequalities affecting ethnic, racial or Indigenous groups; interculturality covers initiatives that explicitly promote

TABLE 6 Studies distribution by publication type.

Publication type	Frequency	%
Journal articles	22	78.6%
Conference papers	6	21.4%

TABLE 7 Classification of reviewed studies by reported technologies and primary technology category.

Study ID	Technologies reported	Assigned technology category
P001	Moodle (forums, blogs for news, wiki for joint curricula work), videoconferences	Learning management and institutional platforms
P002	Video conferencing, Skype, wikis, collaboration platforms	Synchronous and asynchronous communication/collaboration tools
P003	Technology-supported learning environments (TSLE): Internet, pptx, videos, Kahoot	Content-creation and interactive learning tools
P004	Digital School digital platform	Learning management and institutional platforms
P005	Digital Mysteries, Thinking Kit	Content-creation and interactive learning tools
P006	Internet, video creation, presentations	Content-creation and interactive learning tools
P007	Raspberry Pi, App Inventor, robotics	Physical computing, robotics and STEM toolkits
P008	Mobile devices and video creation	Content-creation and interactive learning tools
P009	Virtual learning community (Moodle)	Learning management and institutional platforms
P010	Virtual modality, educational platforms	Learning management and institutional platforms
P011	Video conferencing, teamblogging	Synchronous and asynchronous communication/collaboration tools
P012	Educational robotics, App Inventor, design platforms	Physical computing, robotics and STEM toolkits
P013	Zoom, Padlet, Mentimeter, Jamboard, Innovators Compass framework	Synchronous and asynchronous communication/collaboration tools
P014	Optical laboratories, nanophotonics, AILDI	Physical computing, robotics and STEM toolkits
P015	PowerPoint and projector	Content-creation and interactive learning tools
P016	Virtual tours of museums, web-based quests, virtual newspapers	Immersive and simulation-based learning environments
P017	Digital resources to support STEM activities and virtual training	Physical computing, robotics and STEM toolkits
P018	Teaching simulation in virtual environments	Immersive and simulation-based learning environments
P019	Digital gamification, educational applications	Content-creation and interactive learning tools
P020	Synchronous online collaboration, digital platforms	Synchronous and asynchronous communication/collaboration tools
P021	Metaverse (VR, immersive simulations)	Immersive and simulation-based learning environments
P022	Augmented Reality	Immersive and simulation-based learning environments
P023	Digital learning application (SENANDI)	Content-creation and interactive learning tools
P024	Flipbook, digital media like Canvas, Padlet…	Content-creation and interactive learning tools
P025	BBC micro:bit	Physical computing, robotics and STEM toolkits
P026	Fanfiction, blogs, forums, social networks	Synchronous and asynchronous communication/collaboration tools
P027	AI (OpenAI, MidJourney, Synthesia, DeepAI y Runway), Google Teams, Telegram y Zoom.	Content-creation and interactive learning tools
P028	Videoconference (Zoom)	Synchronous and asynchronous communication/collaboration tools

TABLE 8 Technology categories identified in the studies.

Technology category	Technologies reported in the studies	N° of papers
Learning management and institutional platforms	Moodle (forums, blogs, wiki for joint curricula work), virtual learning community (Moodle), Digital School digital platform, educational platforms, digital platforms, virtual modality	4
Synchronous and asynchronous communication/collaboration tools	Video conferencing, Skype, Zoom, synchronous online collaboration, wikis, collaboration platforms, blogs, forums, social networks, teamblogging	6
Content-creation and interactive learning tools	Internet resources, presentations, PowerPoint, projector, videos, video creation, Kahoot, Digital Mysteries, Thinking Kit, Padlet, Mentimeter, Jamboard, fanfiction, digital gamification, educational applications, AI tools to create content, Digital learning application (SENANDI)	9
Physical computing, robotics and STEM toolkits	Raspberry Pi, App Inventor, robotics, educational robotics, design platforms, optical laboratories, nanophotonics, AILDI, digital resources supporting STEM activities, virtual training	5
Immersive and simulation-based learning environments	Virtual museum tours, web-based quests, virtual newspapers, teaching simulation in virtual environments, Metaverse (VR, immersive simulations)	4

interaction and mutual learning across cultural groups; linguistic diversity addresses multilingualism, language barriers or minoritized languages; and migration refers to mobility-related statuses such as migrants, refugees or international students. When studies addressed multiple dimensions simultaneously, we classified them according to the authors' dominant focus. The near absence of explicit attention to religious diversity in the included studies reflects a limitation of the available empirical base rather than of the scope or design of this review.

The most frequently addressed theme was multicultural awareness in a broad sense ($n = 10$, 35.71%), for example, developing students' social and cultural competencies, working with local cultural diversity, or using CRT approaches. Ethnic and indigenous diversity was the second most frequent dimension ($n = 8$, 28.57%), often in relation to indigenous communities, rural or Native American contexts. Followed by intercultural/cross-cultural diversity ($n = 6$, 21.43%), usually involving structured dialogue or joint projects between groups or schools from different cultural backgrounds. Linguistic diversity was less commonly represented ($n = 3$, 10.71%), typically in the form of bilingual or EFL/CLIL projects that explicitly linked language learning with cultural diversity, while migration-related diversity appeared only once ($n = 1$, 3.57%). No study was coded as having religious diversity as its primary focus, although one cross-border program explicitly addressed Catholic–Protestant divisions in Ireland within a broader intercultural framework. As shown in Table 9, these findings suggest a stronger emphasis on broad multicultural and ethnic diversity issues than on linguistic or migration-related dimensions.

Regarding scope, as shown in Figure 4, most initiatives were implemented in local settings (16 out of 28), such as specific schools, districts, communities or indigenous/rural areas within a single country. Five initiatives were designed or implemented at a national level, understood here as being implemented in more than one site within a single country (for example, national robotics teams or multi-site teacher professional development programs), while seven involved international or cross-border cooperation, connecting classrooms or school groups located in different countries (e.g., Ireland–Northern Ireland, India–UK, Korea–Hawaii, or France–Canada). This suggests that technology-mediated approaches to cultural inclusion in primary and secondary education are still predominantly locally grounded, with a smaller but meaningful subset leveraging digital tools to support cross-national intercultural experiences.

TABLE 9 Distribution of studies by type of inclusion and cultural diversity addressed.

Types inclusion and cultural diversity	Frequency	%
Multicultural awareness (general)	10	35.71%
Ethnic/Indigenous diversity	8	28.57%
Intercultural/Cross-cultural	6	21.43%
Linguistic diversity	3	10.71%
Migration-related diversity	1	3.57%
Total	28	100.00%

3.1.8 Educational levels and participant groups

The six categories were selected to balance conceptual clarity, comparability across very heterogeneous descriptions, and parsimony with the reviewed initiatives.

- Primary (students): Groups all initiatives that target only primary pupils (e.g., "Primary school", "Primary (Pancasila subject)", "Primary school (younger schoolchildren, Grade 3, fourth-grade students"). These cases share similar developmental stages, curriculum context and school organization.
- Secondary (students): Aggregates middle school, junior high, upper-secondary and vocational ("Grade 7 junior high school", "9 grade", "Secondary education", "career & technical high school", etc.), which are all within the broader secondary level and differ substantially from primary in age, subject specialization and school structure.
- Primary and secondary: Used when the intervention explicitly covers a wide grade span crossing the primary–

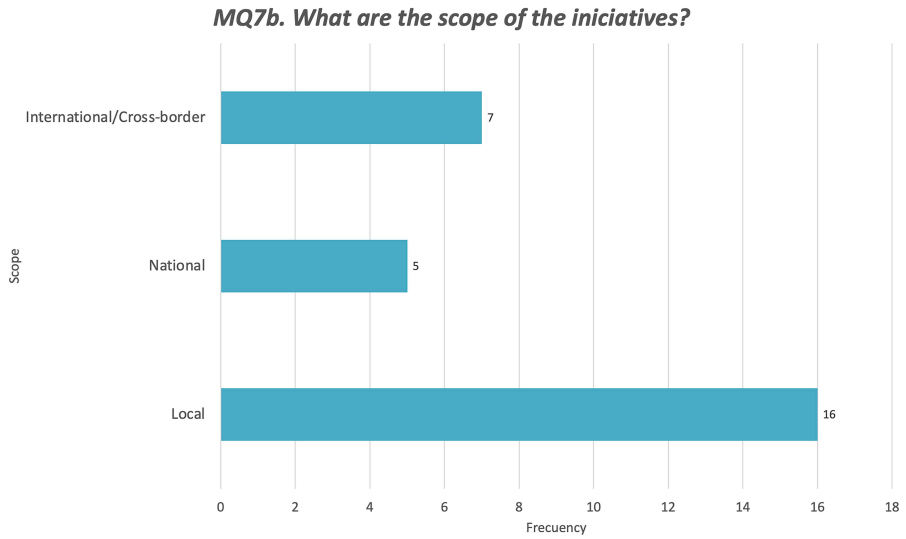


FIGURE 4
Scope of initiatives.

secondary boundary ("primary and post-primary education and special schools", "primary and secondary but for all ages", "grades 3–8", "grades 5–12"). Treating these as a separate group avoids artificially forcing them into primary or secondary.

- Secondary (students and teachers): Capture the case where the study was applied in both groups to avoid forcing them into the students or teachers' category.
- Teacher education: Distinguishes studies focusing on preservice or in-service teachers ("preservice teachers of local primary school", "k12 teachers") from those targeting school students. This is an important concept because outcomes, methods, and implications differ in teacher education.
- All/not specified levels: Captures those cases where the study itself does not specify or explicitly claims "All school levels" or "From school, not age specified", so any finer assignment would be speculative.

Table 10 presents the distribution of the included studies according to educational level and participant group. The studies were conducted mainly at the secondary level, with twelve interventions targeting only middle or upper-secondary students, including general and vocational tracks. One additional study involved both Grade X teachers and students within a secondary school context. Seven studies spanned both primary and secondary education, typically covering wide grade ranges, such as 3–8 or 5–12, or explicitly referring to "primary and post-primary education", and sometimes including special schools. Four interventions focused exclusively on primary education, often working with specific grades (e.g., grade 3 or fourth grade) or subjects such as Pancasila at the primary level. Two studies addressed teacher education rather than pupils directly, involving preservice primary teachers or in-service K–12 teachers participating in culturally responsive or technology-enhanced professional development programs. Finally, two studies reported working across all school levels or did not specify the exact school stage and were therefore coded as "all/unspecified school levels".

TABLE 10 Distribution of studies by educational level and participant group.

Category	*n*	%
Secondary only (students)	12	42.9
Secondary only (students and teachers)	1	3.6
Primary and secondary	7	25.0
Primary only (students)	4	14.3
Teacher education	2	7.1
All/unspecified levels	2	7.1
Total	28	100.0

3.2 RQ1: What types of initiatives are being implemented to promote cultural inclusion and diversity through technology in primary and secondary education?

To answer RQ1, the included initiatives were synthesized according to the predominant dimension of the cultural diversity they addressed. The qualitative synthesis identified five thematic categories following the classification established in the mapping phase (MQ7). The sections describe each category and illustrate it with representative studies.

a. Multicultural awareness

Multicultural awareness refers to initiatives that use technology to help students recognize multiple cultural perspectives and connect curriculum content with local or global cultural contexts. Ten initiatives in Indonesia, Taiwan, Japan, Thailand, New Zealand, United States, Italy and Qatar were classified in this category.

A prominent strategy involves integrating Culturally Responsive Teaching (CRT) with local wisdom to make abstract educational concepts more tangible for students. For instance, the CERITALOGAM model in Indonesia successfully integrates CRT, local Sundanese wisdom, and gamification (using tools such as Kahoot and Wordwall) within Pancasila education to enhance primary students' multicultural competence (Patras et al., 2025). This approach is mirrored in science education, where the integration of ethnopedagogy, utilizing regional Indonesian traditions regarding solar eclipses, has been shown to improve student engagement and cultural identity while simultaneously teaching concepts about the solar system (Rahmawati et al., 2020). Similarly, in Taiwan, a teacher preparation course on Marine Environmental Education utilizes technology-supported environments to help preservice teachers connect sustainable development content with students' local communities and real-life cultural settings (Cheng et al., 2021). This line of work is reinforced by the digital poetry intervention developed in Indonesia, where the SENANDI platform used constructivist pedagogy, TPACK, and culturally responsive teaching to foster nationalism and cultural appreciation among primary students through creative digital writing (Pamungkas et al., 2025).

A second group of initiatives leverages digital collaboration and immersive environments to facilitate direct intercultural interaction or simulated social experiences. The Rete Dialogues network in Italy utilizes team blogging and videoconferencing via the *Generation Global* project to provide an "agentic space" where thousands of students can negotiate their identities and engage in critical dialogue with peers from different regions (Barzanò et al., 2017). This focus on social attitudes is further advanced through metaverse technology in Thailand, where secondary students participate in virtual cultural exploration and role-playing to strengthen their critical thinking and social competence in a simulated multicultural urban environment (Mutchima et al., 2025). Finally, to ensure educators are equipped for these diverse settings, online teaching simulations such as *simSchool* are implemented in the United States to allow high school students interested in teaching careers to practice inclusive strategies with diverse virtual student profiles, significantly increasing their resilience, empathy, and culturally responsive teaching self-efficacy (Christensen and Knezek, 2025).

Other programs focus on STEM as a vehicle for multicultural awareness, framing technical tasks within societal and cultural contexts. In New Zealand, the "Future-Proofing Kiwi Kids" initiative utilizes robotics and Raspberry Pi to help Māori and Pacific students address societal problems, thereby mitigating educational inequalities through culturally safe, collaborative maker spaces (Liesaputra et al., 2020). Similarly, university outreach in Qatar prepares multicultural national robotics teams (including members from Qatar and Afghanistan) for international competitions, fostering a "melting pot pedagogy" in which students from diverse backgrounds develop a sense of diversity through technical collaboration and shared engineering goals (Katbeh et al., 2023). In the same vein, the Japanese Shateki project used micro:bit-based physical computing to situate computational thinking within a familiar festival tradition, showing that culturally meaningful tasks can strengthen student motivation, agency, and belonging while teaching technical content (Sugiyama et al., 2026).

Taken together, these initiatives show that technology is most often mobilized to embed multicultural aims within mainstream subjects and authentic learning tasks, rather than as a stand-alone "diversity" curriculum.

b. Ethnic and indigenous diversity

Ethnic and Indigenous diversity initiatives focus on redressing structural inequalities by strengthening minoritized cultural identities and legitimizing Indigenous knowledge within formal education. Initiatives within this category focus on moving beyond traditional schooling to create educational environments that actively restore cultural identity, challenge structural inequalities, and bridge the gap between traditional knowledge and Indigenous ways of knowing. Eight initiatives in Argentina, Colombia, South Korea, the United States, Russia, and Indonesia exemplify this approach.

A prominent example is the Digital Public Schools (EPD) in Argentina, which serves the Ranquel and Huarpe communities through a personalized, modular ICT system. This shared administration model allows the community to propose local content, such as Indigenous languages, while strengthening individual and social self-esteem (Baldivieso and Carrasco, 2014). This structural approach is mirrored in the United States through Project ExCEED and the ROKET program, which focus on creating culturally relevant STEM pathways for Native American students. Project ExCEED utilizes the CRED framework to situate engineering tasks, such as dam design, within the specific histories and ethics of the Mandan, Hidatsa, and Arikara Nations, while ROKET embeds Indigenous culture and language into optics research to increase student interest and persistence in technical fields (Bowman et al., 2024; Huff-Lohmeier et al., 2017).

Technology is also used to make local wisdom visible through culturally responsive resources in Indonesian multicultural schools. The study on Augmented Reality (AR)-based teaching materials in South Sulawesi showed that embedding local wisdom from Makassar, Bone, Toraja, and Mandar into AR resources significantly improved students' understanding of cultural diversity and local traditions, while also increasing engagement and motivation (Pattaufi et al., 2025). Likewise, the study on CRT and TPACK-based Indonesian language learning tools demonstrated that culturally responsive digital materials can strengthen students' identification with local cultural stories and improve learning outcomes in multicultural school settings (Nisa et al., 2025). Together, these studies show that digital tools are especially effective when they translate Indigenous and local knowledge into concrete, interactive learning experiences.

Technological tools are also utilized to foster ethno-cultural awareness and dismantle racial stereotypes in historically homogeneous societies. In Russia, primary students use virtual tours, museums, and web quests to familiarize themselves with regional indigenous groups such as the Mordvins, Chuvash, and Tatars, involving parents in the step-by-step mastery of ethnocultural ideas (Gubanihina et al., 2017). In the Colombian Caribbean, students use mobile devices to produce low-cost videos to investigate their own cultural legacy. This process promotes intercultural effectiveness by encouraging students to explore their ancestry, thereby laying a foundation for self-

recognition and mutual respect among peers. Both initiatives demonstrate how ICT can transform students from passive receivers of information into active creators of cultural meaning (Quintero Solano et al., 2019). Parallel efforts in South Korea utilize Online Intercultural Exchange (OIE) programs to move beyond superficial textbook coverage, connecting middle schoolers with diverse peers in Hawaii. This approach cultivates critical racial literacy by exposing students to diverse racial realities, such as challenging the stereotype that all Americans have Caucasian features, and by facilitating open discussions about prejudice through platforms like Zoom and Jamboard (Seo and Lim, 2024).

These initiatives empower students to become active members of society by transforming their critical awareness into proactive action plans, thereby strengthening their indigenous identity and preparing them to pursue racial justice in an increasingly multicultural society. Overall, these initiatives illustrate how technology can be harnessed not only to transmit information about ethnic and Indigenous groups but to support collective identity work and critical engagement with racism and colonial legacies.

c. Intercultural and cross-cultural education

Intercultural and cross-cultural education refers to initiatives that use technology to support sustained interaction between learners from different cultural, national, or historical backgrounds, often to bridge physical and social divides. Six initiatives in Ireland/United Kingdom, the United States, France/Canada, Chile, Ukraine and Thailand/Japan were coded under this category.

Programs such as Dissolving Boundaries (Ireland) and the Texas-based Global Projects utilize Moodle, Skype, and videoconferencing to foster empathy and reduce prejudice by engaging K-8 students in joint curricular research—ranging from religious history to butterfly migration (Hopper, 2014; Rickard et al., 2014). The focus on interaction is further refined through experiments between schools in Guadeloupe and Quebec that use synchronous online collaboration on themes such as sustainable development to foster behavioral and emotional skills, including empathy (e.g., tropical sugar cane vs. continental maple syrup) (Parent et al., 2025). Finally, the Intercultural Talent Management Model in Chile utilizes virtual environments (Moodle) to foster collaborative networks among Mapuche communities, reshaping cultural relations, strengthening indigenous identity, and preventing acculturation (Careaga Butter et al., 2015).

A more recent example is a study from Ukraine that used digital technologies and artificial intelligence in foreign-language classes to develop students' multicultural competence through online collaboration, discussion, debate, and AI-supported content creation. The intervention helped students expand their understanding of literary works, strengthen cultural sensitivity, and improve intercultural communication skills, showing that digital environments can support deeper engagement with diverse cultural perspectives even when the learning group is geographically bounded (Ostrovski et al., 2026). Similarly, the SMILE project in Thailand and Japan used internet-based small-group collaboration to connect rural Thai students with Japanese peers in authentic English communication. The project increased confidence, cultural awareness, and willingness to communicate, while also helping students build solidarity and sustain relationships beyond the scheduled sessions (Mayuroh et al., 2025).

These experiences indicate that technology-mediated collaboration can serve as a powerful bridge across geopolitical and cultural divides when it is anchored in shared inquiry and long-term partnerships rather than one-off exchanges.

d. Linguistic Diversity

Linguistic diversity initiatives use digital tools to reposition languages, from both global and ancestral languages, as vehicles for cultural expression, critical literacy and civic participation. Three initiatives in Chile, Colombia and Ecuador belong to this category.

In Chile, the implementation of fanfiction projects within a linguaculture (LC) framework allows students to engage in "bi-linguacultural" practice by reimagining global narratives through their own cultural lens. This pedagogical innovation encourages students to project their identities, for example, by incorporating local Chilean slang into English-language writing, thereby bridging the gap between their native and target linguacultures (Gonzalez-Vidal, 2026).

This institutional approach is mirrored in Colombia through the implementation of Content and Language Integrated Learning (CLIL) projects in state schools, where students from underprivileged backgrounds use technology-enhanced environments to learn environmental science through English. By creating digital artifacts such as e-books and videos, students move beyond simple linguistic acquisition toward scientific citizenship, using a foreign language to investigate local and global ecological issues while reinforcing their own identities (Garzón-Díaz, 2021). In Ecuador, the Ministry of Education implemented the *Mecapacito* virtual platform to provide massive training for teachers in Intercultural Bilingual Education (EIB), aiming to institutionalize ancestral languages like Kichwa and Shuar as official languages of intercultural relations (Estrada et al., 2024).

Collectively, these initiatives suggest that technology can support linguistic justice when it positions students' languages as resources for knowledge production and identity, rather than as isolated school subjects.

e. Migration-Related Dynamics

Migration-related dynamics initiatives engage students with the lived experiences, emotions and complexities surrounding human mobility, often through inquiry-based learning and digital storytelling. One cross-national project between India and England illustrates this strand. Initiatives in this category focus on fostering deep learning by moving beyond memorizing historical facts toward an understanding of the lived experiences and complexities of human mobility.

A central project involved students in India and England utilizing Enquiry-Based Learning (EBL) combined with digital content-creation tools, specifically Thinking Kit and Digital Mysteries, to conduct real-world research. By interviewing community members, such as Syrian immigrants, students

developed a high degree of emotional awareness and empathy, which allowed them to grasp that forced migration is a layered and complex issue rather than a binary one. By investigating the facts and personal experiences associated with human mobility, students engage in deep learning that enables them to unlearn stereotypes and reconstruct their understanding of different countries, fostering a profound sense of cross-cultural empathy (Sarangapani et al., 2019).

This evidence indicates that technology-mediated inquiry can move classroom discussions of migration beyond abstract facts toward critical, affective engagements with contemporary mobility and displacement.

3.3 RQ2: What mechanisms are used to evaluate the impact of these initiatives?

All the initiatives were implemented in real educational contexts rather than remaining at the level of theoretical proposals or simulated activities, providing a solid empirical basis for examining their impact on cultural inclusion and diversity. However, the robustness and depth of the evaluation mechanisms vary considerably across studies. According to the quality criteria, as shown in Table 11, 17 initiatives include a clearly articulated mechanism to assess impact in the short, medium, or long term; 10 provide only partial or indirect evidence; and 1 does not report an explicit impact-evaluation strategy. This initiative is retained because it met the minimum quality threshold and provides a rich description of its design and implementation, although it does not report an explicit impact evaluation.

Evaluation designs are heterogeneous. Mixed-methods approaches predominate (16 out of 28), combining quantitative methods (pre- and post-tests, validated scales) and qualitative methods (interviews, diaries), with simple rubrics, observations and comparisons. Only 7 use rigorous designs (quasi-experimental/longitudinal with control groups), limiting inferences about sustained changes in attitudes or school culture. A smaller number of studies were primarily qualitative, drawing on lesson plans, reflective journals, artefacts, interviews, or ethnographic materials rather than statistical comparisons. In many cases, impact is inferred from short-term changes during a project or course, rather than from systematic monitoring of long-term outcomes for students, teachers or communities.

The following examples illustrate the range of quantitative, qualitative, and technology-mediated evaluation strategies identified:

- Quasi-experimental and longitudinal designs: In Thailand, a metaverse initiative used pre-/post-tests with experimental/control groups and *t*-tests to assess social attitudes and critical thinking (Mutchima et al., 2025). In Ireland, Dissolving Boundaries administered a 43-item questionnaire to participants 1 year later, compared with a non-participant cohort, to examine cross-border diversity (Rickard et al., 2014). In Indonesia, CERITALOGAM measured N-Gain in experimental/control classes (Patras et al., 2025). Similarly, in Indonesia a augmented reality initiative used a quasi-experimental design with 304 students and reported significant gains in cultural awareness and understanding of local wisdom (Pattaufi et al., 2025), while in an initiative in Ukraine developing a course in foreign literature using AI and digital tools used an educational experiment with control and experimental groups and found higher multicultural competence scores in the experimental group (Ostrovski et al., 2026).
- Rubrics and learning analytics: Global Projects in Texas employed a "Global Project Rubric" to score objectives, interaction, cultural diversity, tools, and technology performance (Hopper, 2014). simSchool used Learning Analytics for real-time interactions, with visual benchmarks for teacher self-reflection (Christensen and Knezek, 2025). The SENADI initiative in Indonesia also added application analytics to track user behavior, including login frequency, session duration, and poem production metrics, linking creative output to nationalism gains (Pamungkas et al., 2025).
- Qualitative observation and content analysis: In Taiwan, video recordings of teaching demonstrations were coded by Culturally Responsive Teaching (CRT) constructs (Cheng et al., 2021). In the UK/India, thematic analysis of interviews assessed analytical skills and empathy (Sarangapani et al., 2019). In Colombia, ethnographic diaries and interviews evaluated intercultural negotiation via videos (Quintero Solano et al.. 2019). CLIL in environmental science used teacher journals, interviews, and student artefacts (Garzón-Díaz, 2021). The Indonesian CRT–TPACK study also incorporated needs analysis through classroom observation, interviews, and document analysis, combining taxonomic, componential, and thematic analysis to shape the teaching tools (Nisa et al., 2025).
- Mixed and contextual methods: Project ExCEED (USA) combined validated scales (TESS, CRTSE, CCIS), observations, and focus groups (Bowman et al., 2024). Rete Dialogues (Italy) analyzed 1,500+ pages of digital texts for intercultural curiosity (Barzanò et al., 2017). TEEC (Canada/Guadeloupe) used Design-Based Research with pre-/post-questionnaires and interviews (Parent et al., 2025). In Ecuador/Chile, ethnographic diaries, interviews, and talent/competency models were applied (Estrada et al., 2024). The Japanese micro:bit study on festival traditions used a pilot evaluation with quantitative evidence plus pedagogical interpretation (Sugiyama et al., 2026), while the Thai SMILE project combined questionnaires and interviews to assess confidence, cultural awareness, ICT skills, and collaborative learning (Mayuroh et al., 2025).

This pattern suggests that, while researchers and practitioners are increasingly aware of the need to document the effects of technology-mediated cultural-inclusion initiatives, evaluation remains a weak component of the interventions. Strengthening the use of mixed-methods designs, including longitudinal data and more robust indicators of intercultural learning and inclusion, is a key area for future work. From a school leadership perspective, this limitation is particularly relevant because weak or short-term evidence makes it difficult to justify

TABLE 11 Evaluation mechanism and type of assessment per study.

Paper	Context/Initiative	Evaluation mechanism	Type of assessment
P001	Dissolving Boundaries (Ireland)	Online questionnaire (43 items) to assess attitudes one year after participation.	Quantitative (Longitudinal)
P002	Global Projects (Texas)	A rubric that assesses interaction, diversity and technology and qualitative descriptions.	Mixed
P003	Marine Education (Taiwan)	Video recordings and a coding scheme for five CRT constructs observed every five minutes.	Mixed (Coded observation)
P004	Digital Public School (Argentina)	No evaluation mechanism provided. * Included because it meets the quality threshold (4.5/7), but it contributes descriptive evidence rather than impact evidence.	No assessment provided in the study.
P005	Digital Mysteries (UK/India)	Audio interviews, classroom observation notes and photographs of activities.	Qualitative
P006	CLIL and the Environment (Colombia)	Teacher's journals, questionnaires, interviews and product reviews (e-books, videos).	Mixed (Action Research)
P007	Kiwi Kids (New Zealand)	Questionnaires for students and teachers at the end of the day and open-ended comments,	Mixed with quantitative predominance
P008	Video Production (Colombia)	Cross-cultural effectiveness questionnaires, focus groups and ethnographic field diaries.	Mixed (participatory action research)
P009	Talent Management (Chile)	A closed-ended questionnaire (Likert scale) for teachers and an open-ended survey for students.	Mixed
P010	EIB Training (Ecuador)	Individual and group interviews, and two-column ethnographic field diaries.	Qualitative (Ethnographic)
P011	RD/GG Teamblogging (Italy)	Content analysis of 1,500 pages of digital text and baseline/follow-up questionnaires.	Mixed
P012	Multicultural Robotics (Qatar)	Surveys based on selected criteria (technical skills, communication, goals).	Quantitative
P013	Intercultural Education (Korea/Hawaii)	Zoom recordings, worksheets, interviews and weekly reflections from teachers.	Qualitative
P014	The ROKET Programme (Arizona, USA)	Pre- and post-surveys on attitudes, confidence and teaching skills, as well as open-ended questions and qualitative impact reports	Mixed
P015	Ethnopedagogy in Science (Indonesia)	Reflective journals, interviews, worksheets and a science test	Mixed
P016	Ethnic Awareness (Russia)	Diagnostic experiment, the "Treat candy" sociometric technique, and a creative task.	Mixed (Quasi-experimental)
P017	ExCEED Project (North Dakota)	Validated scales (TESS, CRTSE, CCIS), the COPED observation protocol and focus groups.	Mixed
P018	simSchool (USA)	pre/post validated scales of resilience, empathy and CRT self-efficacy from simulations	Quantitative
P019	CERITALOGAM Model (Indonesia)	N-Gain results (pre/post), expert assessment and satisfaction responses.	Mixed (R&D)
P020	TEEC Project (Canada/Guadeloupe)	Pre- and post-concept questionnaires, Zoom recordings and product analyses.	Mixed (Design-Based Research)
P021	The Metaverse and Learning (Thailand)	Pre- and post-tests on social attitudes and critical thinking, with a *t*-test analysis.	Quantitative (Quasi-experimental)
P022	AR-based teaching materials (Indonesia)	Pre-test and post-test for experimental/control groups, paired *t*-test and independent *t*-test; student perception survey using a 5-point Likert scale.	Quantitative (Quasi-experimental)
P023	Digital Application SENADI (Indonesia)	Expert validation of the SENANDI prototype, pre-test/post-test nationalism attitude scale, application analytics, hierarchical regression, content analysis of student poems.	Mixed (Design-Based Research)
P024	CRT-TPACK-based tools for teaching (Indonesia)	Simplified Borg and Gall model. Needs analysis through observation, interviews and document analysis; expert validation using Gregory's formula; pre-test/post-test with experimental and control groups	Mixed (R&D, quasi-experimental)

(*Continued*)

TABLE 11 Continued

Paper	Context/Initiative	Evaluation mechanism	Type of assessment
P025	Japanese festival game of Shateki (Japan)	Pilot evaluation during a community festival; digital questionnaire using a 5-point Likert scale; confirmation quiz to check understanding of programming concepts.	Quantitative (pilot evaluation)
P026	Fanfiction in EFL (Chile)	Teachers' lesson plans, reflective journals and students' work	Qualitative
P027	Online educational Foreign Literature course (Ukraine)	Educational experiment with experimental/control groups; assessment of multicultural competence through 10 situational tasks on a 12-point scale; student surveys.	Quantitative (Quasi-experimental).
P028	Students Meet Internationally through Language Education (SMILE) project (Thailand/Japan)	Post-course online questionnaire and individual interviews with Thai students after the SMILE project.	Mixed

investments, compare alternatives, and scale technology-mediated cultural inclusion initiatives across school contexts.

3.4 RQ3: What are the main challenges and barriers faced by schools in implementing cultural inclusion and diversity projects?

Across the reviewed studies, schools face a combination of teacher-related, institutional, and contextual barriers when implementing technology-mediated projects on cultural inclusion and diversity. Teachers frequently report limited training and self-efficacy in culturally responsive pedagogy and in integrating technology beyond basic presentation tools, which constrains the depth and consistency of implementation in everyday practice. Time pressure, heavy workloads, and the short duration of pilot projects further limit opportunities to plan, adapt, and consolidate these innovations at the classroom level (Bowman et al., 2024; Cheng et al., 2021; Sarangapani et al., 2019).

At the institutional level, unequal access to devices, connectivity and technical support, especially in rural and socio-economically vulnerable contexts, makes it difficult to sustain online intercultural exchanges or fanfiction and project-based activities that depend on stable platforms. While student-centered projects focused on linguistic diversity demonstrate creative success, large-scale institutional efforts to preserve ancestral languages through digital platforms encounter significant structural and pedagogical hurdles. School policies and infrastructure can also become barriers: restrictive firewalls, licensing and compatibility issues, and strong concerns about student safety and internet monitoring often lead leaders to adopt very conservative approaches to digital tools, reducing space for student-led, critical intercultural work. In some cases, technology labs are reserved for specific subjects, which prevents their use for cross-curricular or community-linked intercultural projects (Gonzalez-Vidal, 2026; Liesaputra et al., 2020; Sarangapani et al., 2019). These barriers are not only technical constraints but also leadership challenges, as they require coordinated decisions about infrastructure, teacher support, risk management, curriculum flexibility, and partnerships.

Curricular and contextual constraints also play a central role. Several studies highlight rigid, exam-driven curricula and language policies (for example, English-only expectations in EFL classrooms) that prioritize measurable language outcomes over critical engagement with culture and identity, leaving little room to integrate ICT-supported intercultural pedagogies. Teachers describe a lack of clear curricular guidance and institutional support for culture-and-technology integration, as well as limited recognition (e.g., in assessment or accountability systems) of the time needed to build partnerships with families, communities, or schools in other regions or countries. Finally, broader social inequalities—such as socio-economic vulnerability, uneven access to quality schooling, and persistent stereotypes about minorities and about who "belongs" in STEM or digital spaces—shape participation, making it harder for culturally diverse students to fully benefit from these initiatives and for schools to sustain them over time (Bowman et al., 2024; Gonzalez-Vidal, 2026; Liesaputra et al., 2020; Parent et al., 2025).

These barriers also interact with the five technological categories identified in MQ6: learning management and institutional platforms are especially sensitive to connectivity, infrastructure and platform access; communication and collaboration tools depend on reliable devices, permissions and scheduling; content-creation and interactive learning tools are constrained by teacher confidence, time and curricular flexibility; physical computing, robotics and STEM toolkits require equipment, technical support and sustained maintenance; and immersive and simulation-based environments are particularly vulnerable to high costs, moderation concerns and limited access to suitable devices.

3.5 RQ4: Which key affordances of digital technologies are used in cultural-inclusion initiatives in primary and secondary education, and in which contexts are they reported?

The evidence indicates that the contribution of digital technologies to cultural-inclusion initiatives lies less in the specific tools themselves than in the pedagogical affordances they enable. Across the reviewed studies, five main groups of technologies were identified, each supporting a different possibility for cultural-inclusion work. They show that different digital environments afford distinct possibilities for implementing cultural inclusion projects. Widely available tools

for content creation, communication and learning management tend to lower entry barriers by fitting into existing classroom routines, whereas more advanced immersive and STEM-oriented technologies open up opportunities for deeper experiential learning but remain less accessible in many contexts.

Learning management and institutional platforms such as Moodle or digital school platforms, afford continuity and coordination. They enable the organization of intercultural projects over time, the hosting of shared resources, and the connection of geographically distant schools under a common curricular framework. Their forum, blog and wiki tools enable asynchronous interaction, allowing teachers and students to contribute when connectivity, time zones or timetables make synchronous meetings difficult, and to embed intercultural activities into regular units rather than as one-off events. In addition, national or regional digital school initiatives provide remote or indigenous communities with access to infrastructure and online resources, a prerequisite for implementing technology-mediated inclusion projects (Careaga Butter et al., 2015; Hopper, 2014).

Synchronous and asynchronous communication/collaboration tools, including videoconferencing, blogs, forums, or wikis, afford intercultural contact across geographical, linguistic, and social distance. Synchronous communication enables real-time dialogue and emotional engagement, while asynchronous collaboration allows students in different time zones, school calendars or linguistic contexts to participate more flexibly. These tools are especially relevant in cross-border or international projects, where cultural inclusion depends on sustained interaction rather than on the simple presentation of cultural information. They make it feasible to sustain dialogue across regions and countries, co-design projects, exchange cultural artefacts and work through sensitive topics such as race, migration or Indigenous rights in real time or with reflective pauses, depending on what is pedagogically and logistically appropriate.

Content-creation and interactive learning tools, such as mobile video, digital storytelling, fanfiction, gamified applications, digital poetry, presentations or AI-supported content creation, afford community-driven knowledge production. They enable students to represent their own histories, languages and identities through multimodal artefacts that can be shared within and beyond the classroom. They become producers of cultural meaning. Because these tools often build on familiar technologies (internet resources, PowerPoint, low-cost devices), they allow teachers to introduce culturally responsive, student-centered tasks without requiring advanced technical expertise, thus supporting scalability.

Physical computing, robotics, and STEM toolkits afford culturally situated problem-solving. Tools such as Raspberry Pi, App Inventor, robotics kits, micro or STEM laboratories make it possible to connect engineering, computing or science learning with local traditions, community challenges and the participation of underrepresented groups. This affordance is especially important because it expands cultural inclusion beyond humanities or language subjects and shows how STEM education can also become a space for identity, belonging and culturally responsive pedagogy.

Finally, immersive and simulation-based learning environments, including virtual tours, augmented reality, metaverse environments and teaching simulations, afford perspective-taking, experiential exploration and safe rehearsal of intercultural situations. These technologies allow students and teachers to explore cultural heritage sites, interact with simulated multicultural scenarios, or practice inclusive pedagogical responses in controlled environments. Despite appearing in fewer studies, these environments illustrate how advanced technologies can support empathy, critical reflection and scenario-based practice in culturally diverse classrooms. From a leadership perspective, these findings suggest that technology choices should be guided by pedagogical affordances, accessibility, teacher readiness, and sustainability, rather than by technological novelty alone.

Table 12 provides a concise synthesis of the main affordance associated with each technology group and one illustrative study from the review.

4 Discussion

4.1 Main findings

The synthesis of the 28 studies allows us to move beyond a descriptive account of individual initiatives and identify three broader contributions to the understanding of technology-mediated cultural inclusion in primary and secondary education.

These findings extend previous reviews on educational technology and inclusion. Whereas earlier reviews have focused on culturally relevant computing education, empathy in STEM, inclusive education more broadly, or school culture, the present review shows that technology-mediated cultural inclusion in primary and secondary education has specific dynamics related to identity recognition, intercultural dialogue, Indigenous knowledge, linguistic diversity, and migration-related experiences (Lunn et al., 2022; Morales-Chicas et al., 2019; Solís-Pinilla et al., 2024). This distinction is important because cultural inclusion cannot be reduced to general inclusion or access to technology. It requires attention to how digital tools mediate representation, dialogue, belonging, and equity in culturally diverse school communities.

First, this review shows that technology-mediated cultural inclusion in schools is not a single type of intervention, but a heterogeneous field organized around different pedagogical purposes. The initiatives reviewed do not simply use technology to address diversity; rather, they mobilize digital tools to make culture visible, to connect learners across social and geographical boundaries, to support culturally responsive curriculum design, and to create spaces for identity negotiation. Across the corpus, technology is most frequently used to embed cultural inclusion within ordinary curriculum areas such as science, language learning, literature, citizenship education and STEM, rather than as a separate diversity-oriented program. This is an important finding because it suggests that cultural inclusion becomes more pedagogically meaningful when it is integrated into mainstream learning tasks and connected to students lived experiences, local knowledge and community identities. For educational leaders, this means that technology-mediated cultural inclusion should be approached not as a tool-selection problem, but as a strategic process involving

TABLE 12 Digital technology affordances for cultural inclusion.

Technology group	Affordance for cultural inclusion	Illustrative study/ initiative
Learning management and institutional platforms	Sustain intercultural or Indigenous education initiatives over time by organizing resources, participation and collaboration in a shared digital space.	P009 – Talent Management
Synchronous and asynchronous communication/collaboration tools	Enable dialogue across geographical, cultural and temporal distance, including flexible participation across different school calendars or time zones.	P011 – RD/GG Teamblogging
Content-creation and interactive learning tools	Support student voice, cultural representation and community-based knowledge production through multimodal artefacts.	P008 – Video Production
Physical computing, robotics and STEM toolkits	Connect technical learning with culturally meaningful problems and broaden participation in STEM among underrepresented groups.	P007 – Kiwi Kids
Immersive and simulation-based environments	Create safe spaces for perspective-taking, cultural exploration and rehearsal of intercultural or inclusive responses.	P021 – The Metaverse and Learning

pedagogical vision, teacher capacity-building, infrastructure planning, and equity-oriented institutional change.

Second, the review clarifies the main forms of technology-mediated cultural inclusion currently in use in school education. The evidence points to five recurring configurations. Learning management systems and institutional platforms support the organization and continuity of intercultural projects; communication and collaboration tools enable contact between geographically distant groups; content-creation and interactive tools allow students to produce cultural meanings through videos, digital stories, fanfiction, poetry or gamified activities; robotics, physical computing and STEM toolkits connect technical learning with culturally meaningful problems; and immersive or simulation-based environments create opportunities for perspective-taking, empathy and scenario-based practice. These configurations show that the educational value of technology does not reside primarily in its novelty, but in the affordances, it provides for participation, representation, collaboration and reflexivity. In this sense, accessible technologies such as videoconferencing, blogs, mobile video, digital storytelling, or collaborative platforms remain central to the field because they can support culturally inclusive practices without requiring highly specialized infrastructure.

Third, the review shows that technology-mediated cultural inclusion is strongly context-dependent. Many initiatives are locally grounded, shaped by specific cultural, linguistic, Indigenous, rural, national or institutional realities. This contextual embeddedness is a strength because it prevents inclusion from being reduced to a generic or universal model. However, it also limits the comparability and scalability of findings. The same digital tool may support intercultural dialogue in one context, linguistic justice in another, and Indigenous knowledge revitalization in another. Therefore, the evidence suggests that technology should be understood as a mediating infrastructure whose inclusive potential depends on how it is pedagogically designed, culturally situated and institutionally supported.

At the same time, several aspects remain uncertain or contested. The first concerns impact. Although most studies report positive outcomes, evaluation remains uneven. Many initiatives rely on short-term, small-scale or self-reported evidence, and only a limited number use quasi-experimental, controlled or longitudinal designs. As a result, the field still lacks robust evidence about whether these initiatives produce sustained changes in intercultural competence, cultural identity, school belonging, teacher practice or institutional culture. The second uncertainty concerns depth of inclusion. Some initiatives promote meaningful intercultural dialogue, critical racial literacy or community-based knowledge production, while others remain closer to cultural awareness or celebratory multiculturalism. This distinction matters because recognizing cultural difference is not the same as challenging structural inequalities, racism, linguistic hierarchies or epistemic marginalization. The third contested issue concerns technology itself. Digital tools can expand access, voice and interaction, but they can also reproduce existing inequalities when schools lack connectivity, devices, technical support, teacher training or curricular flexibility. Thus, technology-mediated inclusion should not be interpreted as inherently inclusive; its effects depend on the social and institutional conditions in which it is implemented. This concern is consistent with broader debates in inclusive education, which warn that inclusion cannot be reduced to participation or access alone, but must address structural inequalities, institutional cultures and equity-oriented change (Ainscow, 2020; Artiles et al., 2011).

Importantly, the confidence that can be placed in these findings varies across studies. The strongest evidence comes from the relatively small number of quasi-experimental and longitudinal investigations, whereas many of the reported benefits originate from exploratory, design-based, qualitative, or short-term studies with limited opportunities for causal inference. Differences in methodological quality are therefore central to the interpretation of the results: studies with control or comparison groups provide more robust evidence of impact, while descriptive or single-site initiatives are more useful for identifying practices, contexts, and implementation conditions. In addition, a few studies reported negative or contradictory findings, which may reflect publication and reporting bias rather than uniformly positive effects. Consequently, the positive outcomes identified in this review should be interpreted as promising rather than conclusive, reinforcing the need for more rigorous evaluations capable of assessing long-term educational and institutional impacts (Page et al., 2021).

These findings have implications for theory. They suggest that technology-mediated cultural inclusion should be conceptualized

at the intersection of multicultural education, intercultural education, culturally responsive pedagogy and digital learning (Banks, 2014; Dervin and Gross, 2016; Gay, 2002; Nieto and Bode, 2012; UNESCO, 2006). From multicultural education, the reviewed initiatives inherit a concern with equity, representation and the transformation of curriculum and school culture. In intercultural education, they emphasize dialogue, reciprocal learning, and interaction across differences. In culturally responsive and culturally relevant pedagogy, they foreground students' cultural backgrounds as resources for learning rather than as deficits to be compensated for (Hammond, 2015). Digital technologies add a further layer: they mediate participation, enable new forms of cultural production, and extend the classroom beyond local boundaries. However, the review also shows that these theoretical traditions are not always explicitly connected in the primary studies. Many initiatives use culturally responsive or intercultural language without fully theorizing how technology changes the conditions under which culture, identity and learning are negotiated.

Accordingly, the main theoretical contribution of this review is to frame technology-mediated cultural inclusion as a relational and socio-technical process. It is relational because inclusion emerges through interactions among students, teachers, communities, curricula and institutions. It is socio-technical because these interactions are shaped by digital infrastructures, platforms, tools and data practices, but are never determined by them. This perspective helps explain why similar technologies lead to different outcomes across contexts and why inclusive aims cannot be achieved through digital access alone. For technology-mediated cultural inclusion to become transformative, digital tools must be embedded in culturally responsive pedagogies, supported by institutional conditions, and oriented toward equity, critical reflection and student agency.

4.2 Limitations

4.2.1 Limitations of initiatives

The initiatives identified in this review show important contributions, but they also present several limitations. First, many initiatives are implemented as short-term or pilot projects, which makes it difficult to evaluate their sustainability over time. In most cases, interventions are limited to a specific course, school year, or context, with little evidence of long-term integration into school systems.

Second, there is a strong reliance on individual teachers or external projects rather than on institutional strategies. This means that the success of many initiatives relies on teacher motivation, specific funding, or temporary collaborations, which limits scalability and replication in other contexts.

Another important limitation concerns technology use. Although technology is central, most initiatives rely on basic and widely available tools (e.g., videos, platforms, videoconferencing). In contrast, more advanced technologies (e.g., VR or immersive environments) are less explored and often remain experimental. This suggests a gap between technological potential and actual classroom practice. This interpretation is consistent with broader inclusive education research, which argues that inclusion requires changes in school cultures, policies and practices, rather than merely adding resources or tools (Ainscow, 2020).

In addition, several initiatives do not fully address structural and contextual inequalities. For example, access to devices, connectivity, and technical support is uneven, especially in rural or vulnerable contexts. As a result, technology-mediated inclusion may unintentionally reproduce existing inequalities rather than reduce them.

Finally, some initiatives show a mismatch between design and implementation. For instance, large-scale teacher-training programs, such as those focused on intercultural bilingual education, may fail when they do not account for local pedagogical needs, cultural contexts, or technical barriers. This highlights that technology alone is not enough to ensure meaningful cultural inclusion. For example, in Ecuador, the Mecapacito virtual platform was designed to provide large-scale teacher training in Intercultural Bilingual Education (EIB), to institutionalize ancestral languages such as Kichwa and Shuar as official languages for intercultural relations. However, research conducted during the 2022–2023 period concluded that these training programs were largely ineffective due to a lack of pedagogical relevance to the specific EIB model and a failure to consider technical barriers and bureaucratic credentialism. These findings highlight a notable discrepancy between the original democratic goals of multiculturalism and their institutional implementation, suggesting that technology-mediated linguistic diversity requires more than just digital access; it demands a radical change in the epistemological structure of the education system to ensure that materials are not merely occidentalized but are deeply rooted in the community's ancestral reality.

4.2.2 Limitations of this review

This systematic review also has some limitations. The selection of studies was restricted to two major databases (Scopus and Web of Science) and to publications in English and Spanish, which may under-represent relevant work published in other languages or indexed in regional databases. The heterogeneity of contexts, methods and outcomes makes it difficult to directly compare effect sizes or to generalize beyond the specific settings studied, and the reliance on peer-reviewed sources means that potentially informative grey-literature reports of less successful initiatives were excluded. In addition, no backward or forward citation tracking or manual searches were conducted. Although this decision enhanced the reproducibility of the review by relying on predefined database searches, it may have resulted in the omission of relevant studies not indexed in Scopus or Web of Science or identified through reference chaining.

Another methodological limitation is that no formal inter-rater reliability statistic was calculated for the screening or eligibility stages. Although three reviewers independently screened the records and disagreements were resolved through discussion until consensus was reached, the absence of a statistic such as Cohen's kappa limits the possibility of quantitatively reporting reviewer agreement.

Finally, because of the considerable methodological heterogeneity of the included studies, no standardized risk-of-bias tool was applied. Instead, methodological quality was

examined through a structured quality assessment checklist tailored to the objectives of the review. Although appropriate for this type of evidence synthesis, this approach does not provide a formal assessment of risk of bias and should therefore be considered when interpreting the findings.

4.3 Research gaps

This review reveals several important methodological and substantive gaps in the literature. First, only 7 out of 28 initiatives used quasi-experimental or controlled designs, and just 1 incorporated any form of longitudinal follow-up, which limits causal inference and evidence about long-term impact. This methodological gap limits the evidence available to school leaders who need to make decisions about investment, scaling, and long-term institutional adoption.

In addition, the literature shows uneven coverage of cultural diversity dimensions: most initiatives focus on general multicultural awareness and ethnic diversity, with ten and eight initiatives, respectively. At the same time, only one initiative in the sample addressed migration-related diversity as its primary focus, none explicitly targeted religious diversity, and no studies examined intersectionality systematically. These patterns show that both methodological rigor and the treatment of underrepresented diversity dimensions remain important blind spots in the literature.

There is also a geographical imbalance, with a stronger presence of studies from the Americas (13) and parts of Asia (12), and less evidence from other regions; Europe is involved in only 6 studies, Oceania in 2 initiatives, and none were found in Africa or Antarctica. Finally, important areas such as teacher education and the use of emerging technologies remain limited (e.g., AI appears in only one study), despite their potential to support more inclusive and transformative practices.

4.4 Bias

Several biases may influence the results of this review. First, publication bias exists because only peer-reviewed studies were included, potentially excluding relevant experiences from the grey literature, where less successful initiatives are more likely to be reported. A language bias is also present, as only studies in English and Spanish were considered, potentially limiting the representation of other cultural contexts.

In addition, database and language biases are likely to have shaped the corpus. Scopus and Web of Science tend to privilege journals published in English and from the Global North, which may lead to an underrepresentation of relevant work from Latin America, Africa or other regions where local journals are not consistently indexed. Restricting the search to English and Spanish further narrows the linguistic scope, making it less likely that studies published in other major languages are captured.

Moreover, many studies rely on self-reported data (e.g., questionnaires or interviews), which may introduce social desirability bias, especially in topics related to inclusion and diversity. There is also a tendency toward positive reporting, with most studies highlighting successful outcomes and offering limited discussion of challenges or failures. Finally, both the primary studies and this review are affected by context and selection biases, as many initiatives are highly context-specific, and the search was limited to specific databases, which may limit the generalizability of the findings.

4.5 Implications

The findings of this review have implications for three main groups of stakeholders: teachers and school leaders, teacher educators and researchers, and policymakers and educational technology designers.

4.5.1 Implications for teachers and school leaders

For teachers and school leaders, the review suggests that technology-mediated cultural inclusion should be integrated into ordinary curriculum planning rather than treated as an isolated or occasional project. Many of the reviewed initiatives were successful because they connected cultural inclusion with mainstream subjects such as science, language learning, literature, citizenship education and STEM. Schools should therefore support teachers in designing activities where students can use digital tools to explore local knowledge, represent their cultural identities, collaborate with peers from different backgrounds and critically engage with diversity. The findings also indicate that accessible technologies can be pedagogically powerful when they are used to promote participation, dialogue and cultural production. Tools such as videoconferencing, blogs, digital storytelling, mobile video, collaborative platforms and gamified applications can support intercultural learning without requiring highly specialized infrastructure. However, their effectiveness depends on pedagogical design. It highlights the importance of sustained professional development that integrates culturally responsive pedagogy with concrete examples of how to use digital tools beyond basic presentation functions. These implications are consistent with research on intercultural competence, which highlights the need for teachers to develop not only knowledge about diversity, but also attitudes, skills and critical awareness for interaction across cultural difference (Council of Europe, 2020). School leaders should therefore provide time, technical support and spaces for collaborative planning, so that teachers can move beyond basic technology use and develop culturally responsive digital practices.

4.5.2 Implications for teacher educators and researchers

For initial and in-service teacher education, the review suggests the need to embed technology-mediated intercultural learning as a core component rather than as an optional add-on, including opportunities for preservice teachers to experience online intercultural exchanges, virtual simulations and community-linked projects themselves. At the same time, researchers should strengthen the evidence base through more longitudinal, mixed-methods and quasi-experimental studies, particularly in relation to underrepresented dimensions such as migration, religious diversity and intersectionality. Future studies should examine how technology-mediated initiatives engage with overlapping forms of

inequality related to language, race, migration status, religion, socio-economic background, rurality and gender.

4.5.3 Implications for policy makers and technology designers

At policy level, investments in connectivity, devices and technical support are necessary but not sufficient; funding schemes also need to recognize the time required to build sustainable partnerships and to adapt resources to Indigenous, minoritized and multilingual communities. Cultural inclusion also requires sustained funding for teacher professional development, curriculum adaptation, community partnerships and long-term evaluation. Policies should therefore avoid treating technology as a stand-alone solution and instead support school-level ecosystems that combine infrastructure, pedagogy and institutional commitment.

Technology designers and ed-tech providers should involve teachers, students and communities in co-design processes to avoid one-size-fits-all solutions that reproduce dominant cultural perspectives. The findings point to the value of platforms and tools that support multilingual content creation, community ownership of data and flexible integration into local curricula.

5 Conclusions

This systematic literature review analyzed 28 technology-mediated initiatives aimed at promoting cultural inclusion and diversity in primary and secondary education. It offers a structured overview of existing work by mapping initiatives across five types of diversity, five technological categories and a range of evaluation approaches, thereby making visible both common patterns and under-represented areas.

Across contexts, the findings show that most initiatives rely on accessible tools and are driven by committed teachers or local projects, but they are often short-term, unevenly evaluated and constrained by infrastructural, curricular and institutional barriers. Important gaps remain regarding migration, religious diversity and intersectional perspectives, as well as in the use of longitudinal and robust evaluation designs.

Future research should use longitudinal, large-scale studies to assess the long-term impact of these initiatives, while also examining underrepresented dimensions of diversity such as migration and religious inclusion. More attention should be given to how emerging technologies, teachers, and institutions can support sustainable, integrated technology-mediated cultural inclusion in education systems.

Overall, this review suggests that effective leadership is a critical condition for transforming isolated technology-mediated cultural inclusion initiatives into sustainable, equitable, and system-wide practices.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: https://doi.org/10.5281/zenodo.20561193.

Author contributions

LG-H: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Visualization, Writing – original draft, Writing – review & editing. FG-P: Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. AG-H: Methodology, Supervision, Validation, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Acknowledgments

This research work has been carried out within the University of Salamanca PhD Program on Education in the Knowledge Society scope (http://knowledgesociety.usal.es).

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. The authors used generative AI tools solely for language editing and translation support. No generative AI tools were used for data analysis, interpretation of results, or manuscript drafting. The authors reviewed and edited all AI-assisted output and take full responsibility for the final content.

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References

Ainscow, M. (2020). Promoting inclusion and equity in education: lessons from international experiences. *Nord. J. Stud. Educ. Policy* 6 (1), 7–16. doi: 10.1080/20020317.2020.1729587

Artiles, A., Kozleski, E., and Waitoller, F. (2011). *Inclusive Education: Examining Equity on Five Continents*. Cambridge, MA: Harvard Education Press.

Baldivieso, S., and Carrasco, S. (2014). Escuela pública digital en pueblos originarios. La propuesta De la universidad De la punta para las comunidades ranquel Y Huarpe De la provincia De San Luis. Argentina. *RED Rev. Educ. Distancia* XIII (41).

Banks, J. A. (2014). *An Introduction to Multicultural Education*, 5th Edn. Boston, MA: Pearson.

Barzanò, G., Cortiana, P., Jamison, I., Lissoni, M., and Raffio, L. (2017). New means and new meanings for multicultural education in a global–Italian context. *Multicult. Educ. Rev.* 9 (3), 145–158. doi: 10.1080/2005615X.2017.1346554

Bowman, F., Klemetsrud, B. J., Ozturk, E., Robinson, J., and Lacina, E. (2024). "Impact of professional development in culturally relevant engineering design for elementary and middle school teachers," in *2024 ASEE Annual Conference & Exposition*. 47561. doi: 10.18260/1-2–47561

Careaga Butter, M., Valenzuela, E. S., and Quintana, M. G. B. (2015). Intercultural talent management model: virtual communities to promote collaborative learning in Indigenous contexts. Teachers' and Students' perceptions. *Comput. Hum. Behav.* 51, 1191–1197. doi: 10.1016/j.chb.2015.01.030

Cheng, M.-M., Lacaste, A. V., Saranza, C., and Chuang, H.-H. (2021). Culturally responsive teaching in technology-supported learning environments in marine education for sustainable development. *Sustainability* 13 (24), 13922. doi: 10.3390/su132413922

Christensen, R., and Knezek, G. (2025). Impact of teaching simulations on resilience, empathy and culturally responsive teaching self-efficacy in career technology teacher preparation students. *Int. J. Technol. Educ.* 8, 104–122. doi: 10.46328/ijte.996

Council of Europe (2020). *Common European Framework of Reference for Languages: Learning, Teaching, Assessment - Companion Volume*. Strasbourg: Council of Europe Publishing.

De Prado, J. E. L., and Gago, A. R. A. (2020). Inclusion through technology and education: a commitment to the improvement of museographic cultural spaces. *Technol. Disabil.* 32 (3), 137–148. doi: 10.3233/TAD-190246

Dervin, F., and Gross, Z. (2016). *Intercultural Competence in Education*. London: Palgrave Macmillan.

Estrada, E., Nataly, J., and Guzñay, J. I. (2024). Multiculturalismo Y capacitación docente en la educación intercultural bilingüE. *Espac. Abierto.* 33 (3), 9–34. doi: 10.5281/zenodo.13119609

García-Peñalvo, F. J. (2022). Desarrollo de estados de la cuestión robustos: revisiones sistemáticas de literatura. *Educ. Knowl. Soc.* 23, e28600. doi: 10.14201/eks.28600

Garzón-Díaz, E. (2021). From cultural awareness to scientific citizenship: implementing content and language integrated learning projects to connect environmental science and English in a state school in Colombia. *Int. J. Biling. Educ. Biling.* 24 (2), 242–259. doi: 10.1080/13670050.2018.1456512

Gay, G. (2002). Preparing for culturally responsive teaching. *J. Teach. Educ.* 53 (2), 106–116. doi: 10.1177/0022487102053002003

Gonzalez-Vidal, T. (2026). Navigating linguacultures through fanfiction: unveiling opportunities and challenges in the Chilean efl school context. *Comput. Assist. Lang. Learn.* 39 (1–2), 89–114. doi: 10.1080/09588221.2024.2366466

Gubanihina, E., Klyueva, E., Kudakova, N., Naumova, T., and Zhestkova, E. (2017). "The pedagogical potential of information and communication technologies in the formation of ethnic and cultural awareness of younger schoolchildren," in *Paper Presented at ENVIRONMENT. TECHNOLOGY. RESOURCES. Proceedings of the International Scientific and Practical Conference*, 06/15.

Gusenbauer, M., and Haddaway, N. R. (2020). Which academic search systems are suitable for systematic reviews or meta-analyses? Evaluating retrieval qualities of google scholar, pubmed, and 26 other resources. *Res. Synth. Methods* 11 (2), 181–217. doi: 10.1002/jrsm.1378

Hammond, Z. (2015). *Culturally Responsive Teaching and the Brain: Promoting Authentic Engagement and Rigor among Culturally and Linguistically Diverse Students*. Thousand Oaks, CA: Corwin Press.

Hoffmann-Eßer, W., Siering, U., Neugebauer, E. A. M., Lampert, U., and Eikermann, M. (2018). Systematic review of current guideline appraisals performed with the appraisal of guidelines for research & evaluation ii instrument-a third of agree ii users apply a cut-off for guideline quality. *J. Clin. Epidemiol.* 95, 120–127. doi: 10.1016/j.jclinepi.2017.12.009

Hopper, S. B. (2014). Bringing the world to the classroom through videoconferencing and project-based learning. *TechTrends* 58 (3), 78–89. doi: 10.1007/s11528-014-0755-4

Huff-Lohmeier, A., Jane, A. J., Hennig, D., and Lamoreaux, M. A. (2017). "Research in optics for K-14 teachers (roket): a research experience for teachers in native American schools".

Katbeh, T., Benjamin Cieslinkski, G., and Bazzi, H. (2023). "Promoting stem education through the preparation of multicultural national robotics teams in Qatar (evaluation)".

Ladson-Billings, G. (1995). Toward a theory of culturally relevant pedagogy. *Am. Educ. Res. J.* 32 (3), 465–491. doi: 10.3102/00028312032003465

Liesaputra, V., Ramirez-Prado, G., Barmada, B., and Song, L. (2020). "Future-Proofing kiwi kids through the use of digital technology," in *Paper Presented at Proceedings of the 51st ACM Technical Symposium on Computer Science Education*, 507–513. doi: 10.1145/3328778.3366902

Lunn, S. J., Bell-Huff, C. L., and LeDoux, J. M. (2022). "Cultivating inclusivity: a systematic literature review on developing empathy for students in stem fields," in *Paper Presented at 2022 CoNECD - Collaborative Network for Engineering and Computing Diversity*. Available online at: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85137173270&partnerID=40&md5=3c902fa1f792af31b2ae0416400e7c8a (Accessed July 9, 2026).

Marinagi, C., Skourlas, C., and Tsolakidis, A. (2017). "Information technology literacy for deaf and hard of hearing students in higher education," in *Strategic Innovative Marketing: 4th IC–SIM, Mykonos, Greece 2015*, eds. A. Kavoura, D. P. Sakas, and P. Tomaras (Cham: Springer International Publishing), 697–703. Available online at: https://ideas.repec.org/h/spr/prbchp/978-3-319-33865-1_85.html (Accessed July 9, 2026).

Martín-Martín, A., Thelwall, M., Orduna-Malea, E., and López-Cózar, E. D. (2021). Google scholar, microsoft academic, scopus, dimensions, web of science, and Opencitations' coci: a multidisciplinary comparison of coverage via citations. *Scientometrics* 126 (1), 871–906. doi: 10.1007/s11192-020-03690-4

Mayuroh, M., Wakabayashi, S., and Iio, J. (2025). The impact of internet-based international small-group activities on students in the countryside. *LEARN J.* 18 (2), 410–429. doi: 10.70730/BJRQ1304

Morales-Chicas, J., Castillo, M., Bernal, I., Ramos, P., and Guzman, B. L. (2019). Computing with relevance and purpose: a review of culturally relevant education in computing. *Int. J. Multicult. Educ.* 21 (1), 125–155. doi: 10.18251/ijme.v21i1.1745

Mutchima, P., Leelakitpaisarn, Y., Pijitkamnerd, B., Phiwma, N., and Pantrakool, S. (2025). The effect of metaverse technology on multicultural learning: strengthening the social attitudes, cultural awareness and critical thinking skills of secondary school students. *J. Educ. e-Learn. Res.* 12 (1), 21–30. doi: 10.20448/jeelr.v12i1.6329

Nieto, S., and Bode, P. (2012). *Affirming Diversity: The Sociopolitical Context of Multicultural Education*. Boston, MA: Pearson Education.

Nisa, K., Syafitri, E., and Simanungkalit, A. (2025). Integrating culturally responsive teaching and tpack frameworks in the design and validation of Indonesian language learning tools for multicultural schools. *Int. J. Learn. Teach. Educ. Res.* 24 (12), 268–294. doi: 10.26803/ijlter.24.12.12

Ostrovski, O., Margitych, K., Siladi, V., Molnar, E., and Udovychenko, L. (2026). Digital technologies and artificial intelligence for multicultural competence in high school literature. *Rev. Eduweb* 20, 303. doi: 10.46502/issn.1856-7576/2026.20.01.18

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., et al. (2021). The prisma 2020 statement: an updated guideline for reporting systematic reviews. *Br. Med. J.* 372, n71 doi: 10.1136/bmj.n71

Pamungkas, O. Y., Sisdiana, E., and Widaryanto, B. (2025). Fostering nationalism through digital poetry: a design-based study integrating constructivist and culturally responsive pedagogy in Indonesian elementary educationeducation. *Res. J. Adv. Humanit.* 6 (4). doi: 10.58256/nwwp3p30

Pandey, P., Jamshidi, F., and Marghitu, D. (2023). "Introducing computer science and arts for all (Csa4all): developing an inclusive curriculum and portal for K5 children. Vol. 14021 LNCS," in *Universal Access in Human-Computer Interaction (Lecture Notes in Computer Science, Vol. 14021)*, eds. M. Antona, and C. Stephanidis (Cham: Springer).

Parent, L., Anjou, C., Chartofylaka, L., De Lacaze, T., Forissier, T., and Bourdeau, J. (2025). Integration of interculturality in education for sustainable development. *J. Environ. Educ.* 56 (3), 246–267. doi: 10.1080/00958964.2025.2471974

Patras, Y. E., Japar, M., Rahmawati, Y., and Hidayat, R. (2025). Integration of culturally responsive teaching approach, local wisdom, and gamification in Pancasila education to develop students' multicultural competence. *Educ. Process Int. J.* 14. e2025045. doi: 10.22521/edupij.2025.14.45

Pattaufi, Makawi, F. E., Aswan, D., and Cahyadi, D. (2025). Developing augmented reality as a teaching material to enhance cultural awareness in secondary schools. *JOIV* 9 (5), 1845–1854. doi: 10.62527/joiv.9.5.4540

Quintero Solano, A., Riveira Zuleta, C. A., Mosquera Arteta, A., Barrios, J. C., and Salcedo, E. M. (2019). Eficacia intercultural y uso de videos: caracterización de la producción audiovisual de estudiantes de básica secundaria del caribe colombiano. *Rev. Colomb. Cienc. Soc.* 10 (1), 119–133. doi: 10.21501/22161201.3063

Rahmawati, Y., Ridwan, A., Cahyana, U., and Wuryaningsih, T. (2020). The integration of ethnopedagogy in science learning to improve student engagement and cultural awareness. *Univers. J. Educ. Res.* 8 (2), 662–671. doi: 10.13189/UJER.2020.080239

Rickard, A., Grace, A. R. C., Austin, R. S., and Smyth, J. M. (2014). Assessing impact of ICT intercultural work. *Int. J. Inf. Commun. Technol. Educ.* 10 (3), 1–18. doi: 10.4018/ijicte.2014070101

Sarangapani, V., Kharrufa, A., Leat, D., and Wright, P. (2019). "Fostering deep learning in cross-cultural education through use of content-creation tools," in *Paper Presented at Proceedings of the 10th Indian Conference on Human-Computer Interaction*.

Seo, Y., and Lim, K. Y. (2024). Racial literacy development in online intercultural education in the efl classroom. *Lang. Cult. Curric.* 37 (3), 400–418. doi: 10.1080/07908318.2024.2349041

Snijder, M., Stapinski, L., Ward, J., Lees, B., Chapman, C., Champion, K., et al. (2021). Strong and deadly futures: co-development of a web-based wellbeing and substance use prevention program for Aboriginal and torres strait islander and non-Aboriginal adolescents. *Int. J. Environ. Res. Public Health* 18 (4), 1–25. doi: 10.3390/ijerph18042176

Solís-Pinilla, J., Miranda-Jaña, C., Merino-Rubilar, C., and Medina-Andrade, J. M. (2024). Educational interactions in school contexts: a systematic review. *Rev. Complut. Educ.* 35 (3), 575–586. doi: 10.5209/rced.85981

Sudirman, A. (2019). Critical issues of teaching innovations for vocational teachers: a literature review. *J. Phys.* 1273 (International Conference on Vocational Education of Mechanical and Automotive Technology 2018), 012038. doi: 10.1088/1742-6596/1273/1/012038

Sugiyama, D., Fahim Ferdous Khan, M., and Sakamura, K. (2026). Integrating Japanese festival traditions into computational thinking: a culturally responsive approach to micro:bit-based physical computing. *Int. J. Adv. Comput. Sci. Appl.* 17 (2), 420–427. doi: 10.14569/IJACSA.2026.0170244

UNESCO (2006). *UNESCO Guidelines on Intercultural Education*. Paris: UNESCO.